Request for Proposal

SELECTION OF TECHNOLOGY PARTNER FOR AI-DRIVEN DOCUMENT VERIFICATION ENGINE UNDER SOCIAL PROTECTION DELIVERY PLATFORM

GOVERNMENT OF ODISHA

ODISHA COMPUTER APPLICATION CENTRE

[TECHNICAL DIRECTORATE OF E&IT DEPARTMENT, GOVERNMENT OF ODISHA]

OCAC Building, Acharya Vihar Square, Bhubaneswar-751013, Odisha, India

W: www.ocac.in | **T**: 0674-2567295/2567283 | **F**: 0674-2567842

Table of Contents

1 Glossary of Terms 7
2 Fact Sheet 8
3 Request for proposal 9
4 Structure of the RFP 9
4.1 Instructions to Bidder 9
4.2 Terms of Reference 9
5 Background Information 10
5.1 Basic Information 10
5.2 Project Background 10
5.2.1 About OCAC 10
5.2.2 Project Profile 11
6 Instruction to the Bidders 11
6.1 General 11
6.2 Compliant Proposals / Completeness of Response 12
6.3 Key Requirements of the Bid 12
6.3.1 Right to Terminate the Process 12
6.3.2 Earnest Money Deposit 12
6.4 Submission of proposal 13
6.4.1 Bid to be submitted in GeM portal 13
6.4.2 Proposal Preparation Costs 14
6.4.3 Language 14
6.4.4 Acceptance and Rejection of Bids 14
6.5 Evaluation Process 14
7 Criteria for Evaluation 15
7.1 Pre-Qualification Criteria 16
7.1.1 General 16
7.2 Technical Evaluation Scoring Matrix 18

7.2.1 Organization Profile [20 Marks] 19
7.2.2 e-Governance Strength [40 Marks] 20
7.2.3 Presentation – Approach and Methodology [40 Marks] 21
7.3 Evaluation of Commercial Bids 23
7.4 Final Evaluation of Bids 24
8 Appointment of System Integrator / Solution provider 24
8.1 Award Criteria 24
8.2 Right to Accept Any Proposal and to Reject Any or All Proposal(s) 24
8.3 Purchaser's Procurement Rights 25
8.4 Notification of Award 25
8.5 Contract Finalization and Award 26
8.6 Performance Guarantee 26
8.7 Signing of Contract 26
8.8 Failure to Agree with the Terms and Conditions of the RFP 27
9 Limitation of Liability 27
10 Terms of Reference (Scope of work) 27
10.1 Background 27
10.2 Project Objective 28
10.3 Challenges in the Current System 29
10.3.1 Delayed Service Delivery 29
10.3.2 High Administrative Burden 29
10.3.3 Error-Prone and Inconsistent Verification 30
10.3.4 Rising Incidents of Document Fraud and Misuse 30
10.3.5 Erosion of Public Trust and System Efficiency 30
10.4 Scope of Work 30
10.5 Overall Application Requirements 34
10.5.1 Web-Based Admin Platform 34
10.5.2 AI-Based Document Verification Engine 35

10.5.3 Document Template Management System 35
10.5.4 Exception Handling & Manual Verification Tools.....35
10.5.5 Administrative Dashboard for Government Officers 35
10.5.6 Integration with Existing Systems 36
10.5.7 Security, Auditability, and Compliance 36
10.6 Technical Requirements:..... 36
10.7 Component Overview and Process Flow 38
10.8 Implementation Strategy 39
10.8.1 Integration with Existing Systems 41
10.8.2 Prerequisites 42
10.8.3 Understand the Applications / Databases.....42
10.9 Integration Methodology – Development of Common APIs 43
10.10 Testing 44
10.11 Cyber Security Audit..... 45
10.12 SSL Certification 45
10.13 Deployment & Configuration 46
10.14 UAT and Go-Live..... 47
10.15 Post Implementation Support..... 48
10.15.1 Application Maintenance 48
10.15.2 System Support.....48
10.15.3 Operation Support 49
10.16 Guiding Principles..... 49
10.16.1 Standards 49
10.16.2 Application 50
10.16.3 Integration 50
10.16.4 Data 50
10.16.5 Data Security.....50
10.16.6 Adherence to Standards 50

10.17 Security, Integrity & Confidentiality 51
10.18 General Terms 52
10.18.1 Change Request Management 52
10.18.2 Exit Plan 53
10.18.3 Project Documentation 53
10.19 Time Line & Tentative Deliverables 54
10.20 Payment Terms 56
10.21 Performance Requirement (SLAs) 57
10.22 Waiver of Penalty 58
10.23 AI related infrastructure 58
10.23.1 Indicative Bill of Quantity (BOQ) 58
10.23.2 Minimum Technical Specifications of Edge AI Runtime Environment 58
10.23.3 Minimum Technical Specifications of Core Hosting Node 61
10.23.4 Technical Specifications of Cloud Management 65
10.24 General Conditions 71
10.25 OCAC Responsibilities 72
11 Formats for Response 73
11.1 Pre-Qualification Bid Formats 73
11.1.1 FORM PQ-1: Cover Letter 73
11.1.2 FORM PQ-2: Bidder’s Organization (General Details) 75
11.1.3 FORM PQ-3: Acceptance of Terms and Conditions 76
11.1.4 FORM PQ-4: Project Citation Format 77
11.1.5 FORM PQ-5: Manufacturer's Authorization Form (MAF) 78
11.1.6 FORM PQ-6: Bank Guarantee Template 79
11.1.7 FORM PQ-7: Format for Non-blacklisting Declaration 81
11.2 Technical Bid Formats 82
11.2.1 FORM TECH-1: Description of Proposed Solution 82
11.2.2 FORM TECH-2: Description of Proposed Approach & Methodology 83

11.2.3 FORM TECH-3: Detailed Work Plan 84
11.2.4 FORM TECH-4: Team Composition, Assignment and Experts' Inputs...85
11.3 Financial Bid 86
11.3.1 FORM FIN-1: Financial Bid Covering Letter 86
11.3.2 FORM FIN-2: Financial Quote 88

1 Glossary of Terms

CIDR	Central Identities Data Repository
DPI	Dots per inch
DSC	Digital Signature Certificate
EMD	Earnest Money Deposit
FRS	Functional Requirement Specification
GST	Goods and Services Tax
ICT	Information and Communication Technology
IT	Information Technology
ITES	Information Technology Enabled Services
KYR	Know Your Resident
Nos	Numbers
OCAC	Odisha Computer Application Center
OSD	Original Software Developer
OSRDH	Odisha State Resident Data Hub
SPDP	Social Protection Delivery Platform
PBG	Performance Bank Guarantee
PMU	Project Management Unit
PSU	Public Sector Undertaking
QCBS	Quality & Cost Based Selection
RFP	Request for Proposal
SSP	Technology Solution Partner

2 Fact Sheet

Sl#	Item	Description
a)	Project Title	Selection of Technology Partner For AI-Driven Document Verification Engine under Social Protection Delivery Platform.
b)	Name of Purchaser	Odisha Computer Application Center (OCAC)
c)	Contact Person, Address and Email	General Manager (Admin) Plot No. N-1/7-D, Acharya Vihar RRL Post Office, Bhubaneswar Odisha - 751013 gm_ocac@ocac.in
d)	RFP Document Fees	Nil
e)	Submission of proposal	The proposals must be submitted online in the portal enivida.odisha.gov.in. Submission of proposals in other forms or portal shall not be considered. For details on submission of proposal in e-Nivida portal. For details, please refer to Clause No. 6.5 of this document.
f)	Earnest Money Deposit	Rs. 14,00,000/- (Rupees Twelve lakhs).
g)	Date and time for Technical Presentation	To be notified later via email
h)	Date and time for opening of Commercial Bids	To be notified later via email
i)	Bid Validity Period	180 days
j)	Project Term	i) Development & Implementation – 4 months from date of work order ii) Post Implementation Maintenance and Support – 36 months from date of Go-Live of the system, which may be extended for another 2 years in case of requirement.

3 Request for proposal

Sealed proposals are invited from eligible, reputed, qualified software application developers and implementers for AI-Driven Document Verification Engine under Social Protection Delivery Platform, Government of Odisha. This invitation to bid is open to all bidders meeting the minimum eligibility criteria as mentioned in this RFP Document.

4 Structure of the RFP

This RFP document for selection of Technology Partner (SSP) for AI-Driven Document Verification Engine under Social Protection Delivery Platform, Government of Odisha comprises of the following parts:

4.1 Instructions to Bidder

The contents of this volume broadly cover following areas:

a) Project Background
b) Instruction to Bidders
c) Criteria for evaluation
d) Appointment of Technology Solution Partner
e) Formats for Pre-Qualification, Technical and Financial Bid response

4.2 Terms of Reference

The contents of this volume broadly cover following areas:

a) About the project and its objectives
b) Scope of work
c) Functional Requirements
d) Project Schedule
e) Payment Terms and Schedule
f) Service level for Technology Solution Partner
g) Timeline of Project implementation
h) Bill of Material and Quantity

5 Background Information

5.1 Basic Information

OCAC, the Technical Directorate of E&IT Department, Government of Odisha invites responses ("Tenders") to this Request for Proposals ("RFP") from Technology Partner firms ("Bidders") for this RFP document for selection of TP for AI-Driven Document Verification Engine under Social Protection Delivery Platform, Government of Odisha as described in the Section-10 "Terms of Reference" of this RFP.

OCAC will award the contract to the successful bidder whose proposal has been determined as the best value proposal based on Technical and Financial evaluation criteria and accepted by the Tender Accepting Authority.

5.2 Project Background

5.2.1 About OCAC

The Department of Electronics & Information Technology is the Nodal Department for Government of Odisha in the matters of IT, ITES and Communication. The department plays a vital role in formulating and implementing policy matters in Information Technology, ITES, Electronics and Telecom; promotion of Odisha as an ultimate ICT destination for investment and facilitating ICT industries; assisting other department for effective e-Governance and capacity building; Promotion of ICT based education in the State.

Odisha Computer Application Centre (OCAC), the Technical Directorate of Electronics and Information Technology Department, Government of Odisha, has evolved through years as a centre of excellence in IT solutions and e-Governance. It has contributed significantly to the steady growth of IT in the state. It helps IT to reach the common citizen so as to narrow down the Digital Divide and widespread applications of IT in establishing a system where the citizens are receiving good governance in addition to ensuring speed of decisions from a transparent Government through an effective e-Governance System.

OCAC on behalf of Government of Odisha intends for AI-driven document verification engine under SPDP. This will be a robust platform with upgraded technology and improved performance, making it more user friendly. OCAC is

pleased to engage qualified and preferred software development agency as Technology Partner to execute this project.

5.2.2 Project Profile

The key objectives of this engagement is as follows.

- **Leverage Artificial Intelligence (AI) and Machine Learning (ML)** to automate the extraction and cross-verification of key details (such as name, date of birth, address, and ID numbers) from various document types.
- Use **pattern recognition and image forensics** to detect anomalies, manipulations, and potential forgeries.
- Support **real-time integration** with official repositories (e.g., DigiLocker, UIDAI e-KYC, academic databases) to validate documents at the point of submission.
- Provide a **standardized verification layer** that can be reused across schemes, reducing redundancy and enabling interoperability across service platforms.
- Reduce dependence on manual verification by enabling officers to act only on cases flagged by the system as exceptions, thus **augmenting human oversight with AI-powered intelligence**.

6 Instruction to the Bidders

6.1 General

a) While efforts have been made to provide comprehensive and accurate background information, requirements and specifications, Bidders must form their own conclusions about the solution needed to meet requirements. Also, bidders may wish to consult their own legal advisers in relation to this RFP.

b) All information supplied by Bidders may be treated as contractually binding on the Bidders, on successful award of the assignment by OCAC on the basis of this RFP.

c) No commitment of any kind, contractual or otherwise shall exist unless and until a formal written contract has been executed by or on behalf of OCAC. Any notification of preferred Bidder status by OCAC shall not give rise to any enforceable rights by the Bidder. OCAC may cancel this public procurement at any time prior to a formal written contract being executed by or on behalf of OCAC.

d) This RFP supersedes and replaces any previous public documentation and communications, and Bidders should place no reliance and dependence on such communications.

6.2 Compliant Proposals / Completeness of Response

a) Bidders are advised to study all instructions, forms, terms, requirements and other information in the RFP documents carefully. Submission of the bid shall be deemed to have been done after careful study and examination of the RFP document with full understanding of its implications.

b) Failure to comply with the requirements of this paragraph may render the Proposal non-compliant and the Proposal may be rejected. Bidders must:

   i) Include all documentation specified in this RFP.

   ii) Follow the format of this RFP and respond to each element in the order as set out in this RFP.

   iii) Comply with all requirements as set out within this RFP.

6.3 Key Requirements of the Bid

6.3.1 Right to Terminate the Process

a) OCAC may terminate the RFP process at any time and without assigning any reason. OCAC makes no commitment, express or implied, that this process will result in a business transaction with anyone.

b) This RFP does not constitute an offer by OCAC. The Bidder's participation in this process may result in OCAC selecting the Bidder to engage towards execution of the contract.

6.3.2 Earnest Money Deposit

a) Bidders shall submit, along with their Bids, EMD of Rs. 14,00,000/- (Rupees Fourteen lakhs) in the shape of Bank Draft OR Bank Guarantee (in the format specified in this RFP) issued by any scheduled bank in favor of Odisha Computer Application Centre" payable at Bhubaneswar and should be valid for 90 days from the due date of the tender / RFP. The EMD should be submitted in the General Bid.

b) EMD of all unsuccessful bidders would be refunded by OCAC within 60 days of the bidder being notified as being unsuccessful. The EMD, for the amount mentioned above, of successful bidder would be returned upon submission of Performance Bank Guarantee.

c) The EMD amount is interest free and will be refundable to the unsuccessful bidders without any accrued interest on it.

d) The bid / proposal submitted without EMD, mentioned above, will be summarily rejected.

e) The EMD may be forfeited:

   i) If a Bidder withdraws the Proposal or increases the quoted prices after opening of the Proposal and during the Bid validity period or its extended period, if any.

   ii) In case, the successful Bidder fails to sign the agreement in accordance with Terms and Conditions (including timelines for execution of the Agreement) of this RFP or fails to furnish Performance Bank Guarantee in accordance with the Terms and Conditions (including timelines for furnishing Performance Bank Guarantee)

   iii) If a Bidder withdraws its bid during the period of bid validity.

   iv) During the Bid process, if a Bidder indulges in any act that would jeopardize or unnecessarily delay the process of bid evaluation and finalization.

   v) If a Bidder has been found to have indulged in any suppression of facts, furnishing of fraudulent statement, misconduct, or other dishonest or other ethically improper activity, in relation to this RFP.

   vi) If a Bidder's proposal contains deviations, conditional offers and partial offers.

f) <u>The local bidders registered under MSME (local MSMEs)/ Startup Odisha are exempted from submission of EMD. However, they must furnish documentary evidence against the same.</u>

6.4 Submission of proposal

6.4.1 Bid to be submitted in GeM portal

6.4.2 Proposal Preparation Costs

The bidder shall be responsible for all costs incurred in connection with participation in the RFP process, including, but not limited to, costs incurred in conduct of informative and other diligence activities, participation in meetings or discussions or presentations, preparation of Proposal, in providing any additional information required by OCAC to facilitate the evaluation process, and in negotiating a definitive contract or all such activities related to the bid process.

OCAC will in no case be responsible or liable for those costs, regardless of the conduct or outcome of the bidding process.

6.4.3 Language

The Proposal should be filled by the Bidder in English language only. If any supporting documents submitted are in any language other than English, translation of the same in English language is to be duly attested by Bidders. For purposes of interpretation of the Proposal, English translation shall govern.

6.4.4 Acceptance and Rejection of Bids

OCAC reserves the right to reject in full or part, any or all bids without assigning any reason thereof. OCAC reserves the right to assess the Bidder's capability and capacity. The decision of OCAC shall be final and binding. Bid should be free of overwriting. All measures, correction or addition must be clearly written both in words and figures and attested. Offers not submitted in prescribed manner or submitted after due date and time are liable to rejection.

6.5 Evaluation Process

a) OCAC will constitute a Proposal Evaluation Committee to evaluate the responses of the bidders.

b) The Proposal Evaluation Committee constituted by OCAC shall evaluate the responses to RFP and all supporting documents/documentary evidence. Inability to submit requisite supporting documents/documentary evidence, may lead to rejection of the bid.

c) The decision of Proposal Evaluation Committee in evaluation of responses to the RFP shall be final. No correspondence will be entertained outside the process of

negotiation/ discussion with the Committee.

d) The Proposal Evaluation Committee may ask for meetings with the Bidders to seek clarifications on their proposals, if required.

e) The Proposal Evaluation Committee reserves the right to reject any or all proposals on the basis of any deviations.

f) Each of the responses shall be evaluated as per the criteria and requirements specified in this RFP.

g) Initial bid scrutiny will be held, and incomplete details as given below will be treated as nonresponsive if proposals are:

   i) Not submitted as specified in the RFP document

   ii) Received without the Letter of Authorization (Power of Attorney)

   iii) Found with suppression of details

   iv) Found with incomplete information, subjective, conditional offers and partial offers submitted

   v) Submitted without the documents requested in checklist

   vi) Submitted with lesser validity period

h) All responsive Bids will be considered for further processing as below:

<u>OCAC will prepare a list of responsive bidders, who comply with all the Terms and Conditions of RFP including specifications of hardware items. All eligible bids will be considered for further evaluation by a committee according to the evaluation process defined in this RFP document. The decision of the Committee will be final in this regard.</u>

7 Criteria for Evaluation

The overall objective of this evaluation process is to select the capable and qualified firm and providing associated capacity building, training and operations and maintenance support.

The Pre-Qualification proposal will be evaluated as per criteria mentioned below and only those bidders who qualify the requirements will be eligible for next level of

evaluations. Technical Proposal and Commercial Proposal of Bidders who do not meet the Pre-Qualification criteria will not be opened in the portal.

The technical score of all the bidders would be calculated as per the criteria mentioned below. All the bidders who achieve at least **70% marks** in the technical evaluation would be eligible for the next stage, i.e. Commercial Bid opening.

Bidders should submit supporting documentary evidence with respect to the above, in absence of which their proposals will be summarily rejected.

7.1 Pre-Qualification Criteria

7.1.1 General

Sl#	Category	Requirement	Compliance
a)	Legal	i) Registered under Companies Act,1956 or a partnership firm registered under the Indian Partnership Act, 1932 or Limited Liability Partnership Act,2008. ii) Company should be in operation for last five (5) years as on date of bid submission date iii) Registered with Goods and Services Tax Network (GSTN).	▪ Copy of Certificate of Incorporation / Registration Or ▪ Copy of the work order or completion certificate as documentary proof of 5 years in operation ▪ Copy of GST Registration Certificate
b)	Turn Over	Average sales turnover of the Bidder from **Software Development and related Service** only (excluding supply of hardware, hardware maintenance, supply of system software and call centre) must be **Rs. 7 Crores** in last three financial years ending on 31st March 2025.	Certificate from CA with Copy of audited Profit & Loss Statement OR Certificate from the Statutory Auditor. The certificate should clearly Startup Odisha Companies must submit required document for same.

Sl#	Category	Requirement	Compliance
		Startup Odisha registered Companies/Local MSEs are exempted from Turnover Criteria	
c)	Net Worth	The net worth of the bidder in last three financial years ending on 31st March 2025 must be positive.	Certificate from the statutory auditor or Internal auditor
d)	Man power	The bidder must have at least 50 full time technical resources (with qualification of M.Tech/B.Tech/ MBA/MCA/MSC(IT)/BSC(IT)) in its payroll as on March 2025.	Copy of the latest EPF deposit challan with HR Declaration with qualification and experience.
e)	Certifications	The Bidder must have CMMI DEV-Level 3 or above (from CMMi) Institute published in CMMi website with validity.	Copy of certificate issued by accredited organizations.
f)	Related Experience	The bidder must have prior experience of development of AI based application which includes Image Processing/ Natural Language Processing (NLP)/ Optical Character Recognition (OCR)/ Speech and Voice Recognition/ Predictive Analytics/Robotic Process Automation	▪ Copy of the Work Order
g)	e-Governance Experience	The bidder should have successfully completed at least following numbers of e-Governance application for any Government Department or Government Agency or PSU in India during last 5 years as on bid submission date and value specified below:	▪ Copy of the Work Order only Software ▪ Project completion/Phase Completion/UAT or go-live certificate.

Sl#	Category	Requirement	Compliance
		i) 1 project not less than ₹2.5 Cr. OR ii) 2 projects not less than ₹2 Cr. each OR iii) 3 projects not less than ₹1.2 Cr. each	
h)	Consortium	Consortium bidding / sub-contracting is not allowed	
i)	Existence in Odisha	The bidder must have a Centre operational in Odisha or submit a declaration to open a office in Odisha within a timeline of 3 months from the award of work.	Trade License/ Leased Agreement
j)	Black listing	The bidder must not be currently under declaration of ineligibility for corrupt and fraudulent practices or blacklisted/debarred by Central Government or any State Government Organization or Department or PSU in India	Self-declaration in this regard by the authorized signatory of the bidder on the company letterhead (as per template provided in this RFP document)
k)	OEM Authorization	The bidder must attach bid specific Manufactures Authorization Form (MAF) letter from OEMs.	Manufacturer's Authorization Form (MAF) (as per template provided in this RFP document)
l)	Earnest Money Deposit	Rs. 14,00,000/- (Rupees Twelve lakhs). Exempted for Startup Odisha registered firms/Local MSEs	In the shape of Bank Draft OR Bank Guarantee (in the format specified in this RFP).

7.2 Technical Evaluation Scoring Matrix

Technical proposal of those bidders will be opened and evaluated, who qualify the Pre-Qualification criteria. The Evaluation Committee will evaluate the Technical Proposals based on technical evaluation criterion as provided below:

7.2.1 Organization Profile [20 Marks]

Sl#	Evaluation Criterion	Max Marks	Compliance
a)	Average sales turnover of the Bidder from Software Development and related Service only (excluding supply of hardware, hardware maintenance, supply of system software and call centre) in last 3 years (2022-2023,2023-2024 & 2024-2025) ≥ 7 crores: 3 marks *[Additional 1 mark for each additional ₹1 crores subject to maximum of 5 marks]* ***For Startup*** ≥ 3 crores: 3 marks *[Additional 1 mark for each additional ₹50 Lakhs subject to maximum of 5 marks]*	5	Copy of audited Profit and Loss Statement **OR** Certificate from the Statutory / Internal Auditor If Startup - Startup Odisha Certificate
b)	The bidder must have at least 50 full-time technical resources in its payroll as on March 2025. ≥ 50 Resources: 3 Mark *[Additional 1 mark for each additional 10 resources subject to maximum 5 marks]*	5	Copy of the latest EPF deposit challan & HR Declaration
c)	**Employee Skills** *1-* PMP/ PgMP *[Each Employee 0.5 marks Maximum 1 marks]* 2- Languages / Technologies	5	Copy of Certification

Sl#	Evaluation Criterion	Max Marks	Compliance
	[Each Employee 0.1 marks Maximum 1 marks] *3-* Database Certification *[Each Employee 0.5 marks Maximum 2 marks]* 4- Security *[Each Employee 0.5marks Maximum 1 marks]*		
d)	The Bidder must have the following valid quality certifications as on bid submission date: CMMi Dev Level 3 or above : 2 Marks ISO 9001 : 1 Marks ISO 27001 : 1 Marks ISO 20000 : 1 Marks	5	

7.2.2 e-Governance Strength [40 Marks]

Sl#	Evaluation Criterion	Max Marks	Compliance
a)	The bidder should have experience in project development of e-Governance project implementation for any Government Department or Government Agency or PSU in India, in the last 5 years as on 31st March 2024. <u>Order value more than ₹1 Crore shall be considered.</u> *[Each project will be awarded 5 marks]*	20	▪ Copy of the work order / Project completion / go-live certificate.

Sl#	Evaluation Criterion	Max Marks	Compliance
b)	Experience in the implementation of AI based Projects covering any of the area – Image Processing – Natural Language Processing (NLP) – Optical Character Recognition (OCR) – Speech and Voice Recognition – Machine Learning – Robotic Process Automation (RPA) Each Project 5 marks	10	▪ Bidder must submit the documentary proof such as Copy of the work order or client certificate or website link etc.
c)	The bidder must have prior experience in API integration with third-party applications as part of web-based integrated e-Governance solutions of Govt. of odisha. *(Each Project with integration with minimum 3 applications shall be considered and each project will be awarded 2.5 marks)*	10	▪ Copy of Work order and documentary proof against Go-live

7.2.3 Presentation – Approach and Methodology [40 Marks]

Sl#	Evaluation Criterion	Max Marks	Compliance
a)	Approach and Methodology a) Understanding of project objectives and challenges – **5 marks** b) Detailed solution design, including AI model lifecycle,	20	Quality of Technical Proposal and Presentation

Sl#	Evaluation Criterion	Max Marks	Compliance
	workflow integration, fallback handling and Implementation strategy including pilot, scale-up, and support – **5 marks** c) Ethical AI use, compliance (IT Act, Aadhaar Act, DPDP), and model transparency – **5 marks** d) Strategy for continuous model improvement, feedback loops, and performance tuning – **5 marks**		
b)	a) Proposed Solution and Its Components with Proposed Architecture for Deploying AI Models on Government Cloud/On-Prem Infrastructure, Optimal Sizing of Required GPU/Compute Resources, and Environment Setup Plan– **10 marks** b) Prototype Demonstration– **10 marks**	20	Evaluation based on Technical Proposal submitted, Presentation & Prototype Demonstration by the Bidder

a) All the bidders who secure a Technical Score of more than **70%** will be declared as technically qualified.

b) The bidder with highest technical bid (H1) will be awarded 100% score.

c) Technical Scores for other than H1 bidders will be evaluated using the following formula:

   Tn = {(Technical Bid score of the Bidder/ Highest technical evaluation marks * 100} % (Adjusted to two decimal places)

d) The commercial bids of only the technically qualified bidders will be opened for further processing.

e) **The bidder must provide a Manufacturer's Authorization Form (MAF) from the Original Equipment Manufacturer (OEM) for all quoted hardware and system**

software items

f) **The bidder must comply with all specifications of hardware and system software items mentioned in the RFP document.**

g) **Bids will be disqualified if the bidder fails to meet the specified hardware requirements, even if they have scored more than 70 marks in the technical evaluation**

7.3 Evaluation of Commercial Bids

a) The Commercial Bids of technically qualified bidders (i.e. Bidders with more than 70 marks in Technical Evaluation) will be opened on the prescribed date in the presence of bidder representatives.

b) Only fixed price financial bids indicating total price for all the deliverables and services specified in this bid document will be considered.

c) Any conditional bid would be rejected.

d) Errors & Rectification: Arithmetical errors will be rectified on the following basis: "If there is a discrepancy between the unit price and total price that is obtained by multiplying the unit price and quantity, the unit price shall prevail and total price shall be corrected. If there is a discrepancy between words and figures, the amount in words will prevail. If the bidder does not accept the correction of error, its bid will be rejected".

e) If there is no price quoted for certain material or service, the bid shall be declared as disqualified.

f) In the event that there are two or more bidders having the same value in commercial bid, the bidder securing highest technical score will be adjudicated as "Best responsive bid" for award of the Project.

g) The bidder with lowest qualifying financial bid (L1) will be awarded 100% score. Financial score for other bidders will be evaluated using the following formula: Fn = {(Financial Bid of L1 / Financial Bid of Bidder) * 100} %.

h) Abnormally Low Bid (ALB) :

A financial bid shall be considered abnormally low if (a) it is more than 20% below the average of all responsive bids, or (b) it is less than (Mean – 1 Standard Deviation) of all responsive bid prices, where:

$$\text{Mean} = \frac{\sum_{i=1}^{n} P_i}{n} \quad ; \quad \text{Standard Deviation} = \sqrt{\frac{\sum_{i=1}^{n}(P_i - \text{Mean})^2}{n}}$$

(where 'P_i ' is the price of the i-th responsive bid and 'n' is the total number of responsive bids).

If a bid is identified as abnormally low, the bidder shall provide a detailed price justification including staffing, overheads, methodology, and profit margins. If the justification is unsatisfactory and the bid is deemed unrealistic or infeasible for proper execution of consultancy services, the bid may be rejected. No automatic additional performance security shall be imposed solely due to the bid being low, except with prior approval of the competent authority.

7.4 Final Evaluation of Bids

a) The technical and financial evaluation scores secured by each bidder will be added using weightages of **70% and 30%** respectively to compute composite score. The composite score will be computed as under:

b) Bn = 70%*Tn + 30%*Fn

c) The bidder securing highest composite score will be adjudicated as most responsive bidder for award of project.

8 Appointment of System Integrator / Solution provider

8.1 Award Criteria

OCAC will award the Contract to the successful bidder whose proposal has scored the highest composite score and would consider it as substantially responsive as per the process outlined above.

8.2 Right to Accept Any Proposal and to Reject Any or All Proposal(s)

OCAC reserves the right to accept or reject any proposal, and to annul the tendering process/ public procurement process and reject all proposals at any time prior to award of contract, without thereby incurring any liability to the affected bidder or

bidders or any obligation to inform the affected bidder or bidders of the grounds for OCAC action.

8.3 Purchaser's Procurement Rights

Without incurring any liability, whatsoever to the affected bidder or bidders, the Purchaser reserves the right to:

a) Amend, modify, or cancel this tender and to reject any or all proposals without assigning any reason.
b) Change any of the scheduled dates stated in this tender.
c) Reject proposals that fail to meet the tender requirements.
d) Exclude any of the module(s)
e) Remove any of the items at the time of placement of order.
f) Increase or decrease no. of resources supplied under this project.
g) Should the Purchaser be unsuccessful in negotiating a contract with the selected bidder, the Purchaser will begin contract negotiations with the next best value bidder in order to serve the best interest.
h) Make typographical correction or correct computational errors to proposals
i) Request bidders to clarify their proposal

8.4 Notification of Award

Prior to the expiration of the proposal validity period, OCAC will notify the successful bidder in writing or by fax or email, that its proposal has been accepted. In case the tendering process/public procurement process has not been completed within the stipulated period, OCAC may like to request the bidders to extend the validity period of the bid.

The notification of award will constitute formation of the Contract. Upon the successful bidder's furnishing of Performance Bank Guarantee (PBG), OCAC will notify each unsuccessful bidder and return their EMD.

8.5 Contract Finalization and Award

OCAC may also like to reduce or increase the quantity of any item in the Scope of Work defined in RFP. Accordingly, total contract value may change on the basis of rates defined in the financial proposal.

8.6 Performance Guarantee

a) OCAC will require the selected bidder to provide a Performance Bank Guarantee (PBG), within 15 days from the date of notification of award.

b) PBG would be **5% of the cost bid value excluding GST** and should be valid for **40 months**.

c) The selected bidder shall be responsible for extending the validity date and claim period of the Performance Guarantee as and when it is due on account of non-completion of the service during the work order period.

d) In case the selected bidder fails to submit performance guarantee within the time stipulated, OCAC at its discretion may cancel the order placed on the selected bidder and/or forfeit the EMD after giving prior written notice to rectify the same.

e) OCAC shall invoke the performance guarantee in case the selected bidder fails to discharge their contractual obligations during the period or OCAC incurs any damages due to bidder's negligence in carrying out the project implementation as per the agreed terms & conditions.

f) <u>The local MSE bidders registered under MSME (local MSMEs)/Startup Odisha Registered firms would be required to make payment of 25% of the prescribed value of Performance Security so as to ensure the performance of the contract (reference Finance Department, Government of Odisha, Office Memorandum No. 13681/F, dated 19.04.2018).</u>

8.7 Signing of Contract

After OCAC notifies the successful bidder that its proposal has been accepted, OCAC shall enter into a contract with the successful bidder (prime bidder in case of consortium), incorporating all clauses, pre-bid clarifications and proposal of the bidder.

8.8 Failure to Agree with the Terms and Conditions of the RFP

Failure of the successful bidder to agree with the draft legal agreement and Terms & Conditions of the RFP shall constitute sufficient grounds for the annulment of award, in which event OCAC may call for new proposals from the interested bidders. In such a case, OCAC shall invoke the PBG of successful bidder.

9 Limitation of Liability

Except in cases of gross negligence or wilful misconduct: -

9.1 neither party shall be liable to the other party for any indirect or consequential loss or damage, loss of use, loss of production, or loss of profits or interest costs, provided that this exclusion shall not apply to any obligation of the supplier/ selected bidder to pay liquidated damages to the Purchaser; and

9.2 Maximum liability of the bidder for this project will be limited to the total value of the contract and will not include any indirect or consequential clause or damage, loss or profit, data or revenue**.**

10 Terms of Reference (Scope of work)

10.1 Background

The Government of Odisha has taken significant strides in digitizing citizen services through portals like **Odisha State Scholarship Portal**, **e-District**, **SUGAM**, **Krushak Odisha**, and several departmental schemes. Despite these advancements, one of the most persistent and time-consuming bottlenecks in the citizen service lifecycle is **manual document verification**. This step is often carried out by human operators at district, block, or municipal levels, resulting in delayed approvals, inconsistent validations, and human errors.

Currently, citizens are required to upload supporting documents (such as Aadhaar cards, caste certificates, income proofs, etc.) while applying for services online. These documents are then manually examined by government staff for authenticity, completeness, and correctness. However, with thousands of applications being submitted every day across multiple departments, this manual verification process is becoming increasingly unmanageable.

Additionally, the **format diversity** in documents (e.g., different versions of Aadhaar or Ration cards, regional language prints, handwritten affidavits, etc.) makes it even more difficult for officials to ensure consistent evaluation. Often, documents that are tampered with or photoshopped go unnoticed due to visual similarities, while genuine documents may get rejected due to unclear scans or minor errors.

To address these challenges, the government has envisioned an AI based solution to assist departments in verifying scanned documents automatically and accurately, reducing dependency on manual checking.

By integrating AI into the backend and simplifying the frontend interactions, this solution aims to not only enhance operational efficiency but also uphold the core tenets of **digital inclusivity, service accessibility, and good governance**.

10.2 Project Objective

To address the challenges comprehensively, there is a compelling need for an **automated, intelligent, and scalable solution** that can streamline the document verification process, reduce administrative workload, and significantly enhance accuracy, fraud detection, and turnaround time.
A modern solution should:

- Leverage OCR for Converting paper documents to digital format for easier storage and access. Also to automate the process of extracting information from images and other documents.
- **Leverage Artificial Intelligence (AI) and Machine Learning (ML)** to cross-verify key details (such as name, date of birth, address, and ID numbers) against different reference system, including source systems.
- Use **pattern recognition and image forensics** to detect anomalies, manipulations, and potential forgeries.
- Support **real-time integration** with other repositories (e.g., DigiLocker, UIDAI e-KYC, academic databases) to validate documents at the point of submission.
- Provide a **standardized verification layer** that can be reused across schemes/systems, reducing redundancy and enabling interoperability across service platforms.

- Reduce dependence on manual verification by enabling officers to act only on cases flagged by the system as exceptions, thus **augmenting human oversight with AI-powered intelligence**.

10.3 Challenges in the Current System

While existing multi-tiered manual verification process is designed to ensure integrity, it has become a bottleneck in the timely and effective delivery of services. The process is characterized by several systemic challenges:

10.3.1 Delayed Service Delivery

While Odisha has made commendable strides in digitizing public services through platforms like e-District and Go Sugam, service delivery timelines continue to suffer due to the absence of automation in the verification process. The current system still relies heavily on manual scrutiny of uploaded documents, leading to delays when staff availability is low or documents are not in the expected format. Citizens often face repeated requests to re-upload documents due to issues like blurriness, cropping, or naming mismatches, prolonging the service lifecycle unnecessarily. For critical schemes such as scholarships, rations, or farmer benefits, these delays translate into real-life hardships and erode the perceived efficiency of digital governance.

10.3.2 High Administrative Burden

Government departments remain under significant administrative load, allocating skilled manpower for repetitive and manual tasks such as verifying Aadhaar cards, income proofs, and supporting certificates. A single document, like an Aadhaar card, may be verified multiple times across different schemes, wasting both time and effort. Employees who could otherwise be addressing complex citizen queries or monitoring service delivery are forced to spend hours each day on mundane document checking. This inefficiency becomes more pronounced during peak application periods like exam result season, ration renewals, or new scheme launches, resulting in slower service across the board.

10.3.3 Error-Prone and Inconsistent Verification

Due to human involvement in verifying documents across various districts and departments, the process lacks uniformity and is subject to interpretation errors. What passes verification in one block may be rejected in another, often based on visual clarity, minor discrepancies in names, or misalignment of scanned documents. This inconsistency leads to unnecessary citizen frustration and contributes to high resubmission and rejection rates. Moreover, because the decision trail is often undocumented, citizens have no insight into why their documents failed validation, hampering transparency and accountability.

10.3.4 Rising Incidents of Document Fraud and Misuse

With more services moving online, there has been a noticeable rise in attempts to manipulate or forge documents to gain unlawful access to benefits. Cases of photoshopped Aadhaar cards, tampered caste or income certificates, and identity fraud using someone else's credentials have increased. Manual verification teams are ill-equipped to catch sophisticated digital forgeries, especially when fake documents closely resemble genuine ones. The absence of automated document authenticity checks or pattern recognition tools further increases the risk of undetected fraud, compromising the integrity of welfare schemes.

10.3.5 Erosion of Public Trust and System Efficiency

When honest citizens experience frequent delays or rejections due to procedural inefficiencies, and fraudsters manage to bypass checks, it leads to a growing distrust in the digital governance ecosystem. Citizens begin to prefer physical visits or agent-based intermediaries over online platforms, which defeats the purpose of e-governance initiatives. Rising grievances and appeals also overburden the support infrastructure, draining time and resources. If not addressed through intelligent automation and consistent rules, the credibility of government platforms will diminish, affecting adoption rates and future digital transformation goals.

10.4 Scope of Work

The system should be capable of verifying a wide range of personal and official documents commonly required for government services in Odisha. This includes, but is not limited to:

- *Identity and Address Proofs*: **Aadhaar Card**, **PAN Card**, **Voter ID**, **Passport**, **Driving License** (DL), etc.

- *Residential, Caste, and Income Certificates*: **Residence Certificate**, **Caste Certificate**, **Income Certificate**, **Solvency Certificate** and other proof-of-entitlement documents issued by competent authorities.
- *Civil Registration Documents*: **Birth Certificate**, **Death Certificate**, **Marriage Certificate** and similar records.
- *Welfare and Financial Documents*: **Ration Card**, **Bank Passbook/Statements** (for account verification), **Job Card** (e.g., MGNREGA job card), Labour Card etc.
- *Educational Certificates*: **Mark sheets**, **Degree Certificates**, **Board examination certificates**, **University diplomas**, etc., which are often needed for scholarships or job applications or admissions in educational institutes.

The AI system will employ **Optical Character Recognition (OCR)** to extract text from scanned copies or photos of these documents. It will then apply **machine learning models** and rule-based checks to verify authenticity and consistency of the information. Key capabilities envisioned are:

- **Data Extraction and Cross-Verification**: Automatically read key fields (names, dates, unique IDs, etc.) from each document and cross-verify them against the application data or other submitted documents. For instance, the name on an income certificate should match the name on the Aadhaar card; any mismatch would be flagged for review. Similarly, an Aadhaar number or PAN number can be validated for correct format and existence (via available government APIs or offline verification tools).
- **Fraud Detection**: Utilize AI to detect forged or tampered documents. This could include checking for visual signs of manipulation (e.g., inconsistent fonts or layout, erased and reprinted text in scanned images) and leveraging databases of known valid document templates. Machine learning models can be trained on genuine and fake document images to identify subtle differences. AI-based anomaly detection can flag when a submitted document deviates from normal patterns (for example, an unusually low-quality scan of a critical field, or a document type that is not usually required for that service). Over time, the system “learns” from known fraud cases to improve its alerts for new attempts.

- **QR Code Detection, Verification, and API Integration**

  To strengthen document authenticity verification, the selected System Integrator (SI) shall develop an **AI/ML-based solution** for QR code detection, scanning, and genuineness verification. The solution shall include:

  - **QR Code Detection and Decoding**: Automatically detect and extract QR codes from scanned documents or uploaded images across various formats (PDF, JPEG, PNG). Decode standard and secure QR codes to retrieve embedded information.
  - **Genuineness Verification through API Consumption**: Connect to and consume APIs provided by the respective issuing authorities (e.g., Education Boards, DigiLocker, e-District, etc.) to verify the authenticity of the QR code data in real time.
  - **API Development and Integration**:
    - Where necessary, develop secure, standardized APIs to interface between the QR module and backend systems that lack ready APIs.
    - Ensure seamless integration of QR verification results into the broader AI-based document verification workflow, with real-time status updates (e.g., "QR Verified", "Verification Failed", "QR Not Found").
  - **Fallback and Exception Management**: Handle cases where QR codes are absent, unreadable, or when API responses are delayed or unavailable, by flagging such instances for manual verification.
  - **Security and Compliance**: Implement secure data transmission (encryption, authentication) and maintain audit logs in compliance with government cybersecurity and data protection standards.
  - **Scalability and Extensibility**: Design the system to easily onboard new document types and integrate additional verification APIs as more certificates incorporate QR codes in the future.

- **Relevance and Completeness Checks**: Ensure the citizen has submitted the **correct and complete set of documents** for the service. The AI can match the document type against a list of required documents for that service request.

If irrelevant documents are uploaded (for example, a **driving license** is uploaded in place of a **caste certificate**), the system will flag it as an incorrect submission. If any required document is missing, it can prompt for the same.

By deploying these capabilities, the government can **significantly speed up the verification process**. AI operates in real-time or near-real-time, meaning that what used to take days of shuttling files between desks could be done in seconds by the system. AI-based document verification system will bring the following benefits:

- **Faster Processing & Reduced Backlogs**: Automated checks happen in seconds, enabling instant verification of most documents and speeding up service delivery.
- **Improved Accuracy & Fraud Prevention**: Machine accuracy in reading data and cross-checking can outperform manual checks. AI can catch forged or tampered documents with a high success rate, reducing incidences of fraud that manual processes fail to catch
- **Consistency in Verification**: The same standards are applied to all applications, eliminating human subjectivity. This ensures equal treatment and reduces errors.
- **Lower Workload on Officials**: Routine verification tasks are handled by the system, freeing government staff to focus on complex cases and other responsibilities. Officials would intervene only for exceptional cases flagged by the system, or to make final approvals, thereby **augmenting human decision-making with AI insights**.
- **Audit Trail and Analytics**: Every check performed by the AI can be logged, creating a digital audit trail. Analytics dashboards can be built to identify common failure points (e.g., which document type sees the most mismatches) and to continuously improve service guidelines.

The selected solution should preferably use open standards and be interoperable with national digital public goods like DigiLocker, etc. in line with the Government of India's **Digital India** and **India Stack** approach to e-governance. To meet this expectation, it is desired to include:

1. **Multilingual OCR Support**: Explicitly require OCR support for Odia and other regional languages with high accuracy. For example, "The OCR engine must

support Odia, Hindi, and English with ≥95% accuracy for printed text and ≥85% for handwritten text."

2. **Document Format Flexibility**: Specify support for diverse document formats, including low-quality images and non-standard layouts. For example, "The system must process documents in JPEG, PNG, PDF, and TIFF formats, including low-quality scans (Minimum 100 DPI) and non-standard layouts."
3. **Advanced Fraud Detection**: Include specific fraud detection techniques, such as deep learning-based image forensics or watermark verification. For example, "The system shall use deep learning to detect tampering (e.g., font inconsistencies, altered pixels) and verify watermarks or holograms."
4. **Citizen Guidance Features**: Add features to guide citizens during document upload, such as real-time image quality checks. For example, "The system shall provide real-time feedback on document uploads, alerting citizens to issues like blurriness or incorrect document types before submission."

10.5 Overall Application Requirements

10.5.1 **Web-Based Admin Platform**

- **Live AI Verification Dashboard: The** web app will display real-time outcomes of OCR confidence scores, AI-based document verification status, exception reports, and document-level audit trails. system health metrics and flag entries requiring manual intervention. This will enable government officers to monitor and audit the verification pipeline effectively.
- **User Role Segmentation and Secure Access Control**: Integrated role-based access control to assign specific permissions to different user categories like data entry operators, verifiers, and supervisors.
- **Document Template Training Interface:** Admin users to upload new document layouts (e.g., updated formats of certificates) into the system and train AI models to recognize them. An intuitive interface allows tagging key fields, bounding regions, and training the system to recognize new templates.
- **Workflow Configuration:** Allow configuration of workflows, AI threshold scores, and override conditions through an intuitive admin console.
- **Real-Time Analytics**: The system will provide visual insights into number of applications processed, error rates, document rejections, and verification predictive trends for fraud detection. Reports can be downloaded or scheduled periodically for review by decision-makers.
- **Exception Handling with Manual Review:** In cases where document quality is low or the AI is uncertain, those entries are flagged for manual intervention.

Officers can override, reject, or approve such cases with full traceability and audit logging.

10.5.2 **AI-Based Document Verification Engine**

- Build an AI engine that performs image classification and pre-processing on uploaded documents before initiating verification.
- Support verification of multiple versions of official documents (e.g., Aadhaar 2013, 2017, 2021) by matching them against trained templates.
- Use OCR to extract structured data like name, ID number, and date of birth from the document.
- Implement multi-level authenticity checks including:
    - Structural template validation
    - Detection of government seals and official signatures
    - QR code or barcode reading for embedded metadata
- Assign AI-generated confidence scores to each verification step and flag low-confidence cases for human review.
- Enable continuous learning from new data to enhance accuracy and reduce false negatives over time.

10.5.3 **Document Template Management System**

- Provide a web interface for authorized officers to upload new formats of documents that citizens may submit.
- Let administrators define bounding boxes and key-value fields on sample documents to train the AI template engine.
- Include a validation preview feature where users can test document formats and see how the system interprets them before deployment.

10.5.4 **Exception Handling & Manual Verification Tools**

- Automatically flag low-confidence or unclear document uploads for manual inspection through the admin panel.
- Provide annotation tools for officers to mark issues (e.g., "blurred text", "signature missing") and approve/reject entries accordingly.
- Maintain full audit logs of manual interventions for accountability and traceability in verification processes.

10.5.5 **Administrative Dashboard for Government Officers**

- Implement a secure web-based dashboard for tracking verifications, managing document formats, and monitoring AI performance.

- Allow manual review of flagged documents and generate MIS reports and performance analytics.
- Enable rule configuration per service type (e.g., certificate expiry limits, mandatory fields).
- Enable filtering by district, document type, date range, or officer for granular insights.

10.5.6 **Integration with Existing Systems**

- Provide secure APIs to integrate the AI verification layer and mobile app with platforms like SPDP, e-District, SAMS, Go Sugam, and others.
- Ensure smooth exchange of data (inputs and results) with issuing authorities and citizen service portals.
- Maintain data localization by ensuring all transactions remain within state-approved infrastructure.

10.5.7 **Security, Auditability, and Compliance**

- Ensure the system complies with IT Act, Aadhaar Act, DPDP Act, and applicable MeitY/CERT-In guidelines.
- Implement access logs, audit trails, role-based access control, and periodic security assessments.
- Guarantee that no citizen data is processed or stored outside Indian jurisdiction.

10.6 Technical Requirements:

1. In compliance with the applicable legal frameworks including the Information Technology (IT) Act, 2000, the Aadhaar (Targeted Delivery of Financial and Other Subsidies, Benefits and Services) Act, 2016, and the Digital Personal Data Protection (DPDP) Act, 2023, it is mandatory that all citizen data and associated processing activities remain within the sovereign jurisdiction of India. Accordingly, all AI/ML-based solutions proposed under this project must strictly adhere to the following:

    a. **On-Premises Deployment of AI Models:** The bidder shall design, develop, and deploy the AI-based document verification system entirely on-premises, within the secure infrastructure provided by the State Data Centre (SDC) or any other government-notified hosting facility located within India. Use of third-party, cloud-based AI platforms (such as OpenAI,

Gemini, Copilot, or similar) that transmit or process data outside the boundaries of India is strictly prohibited.

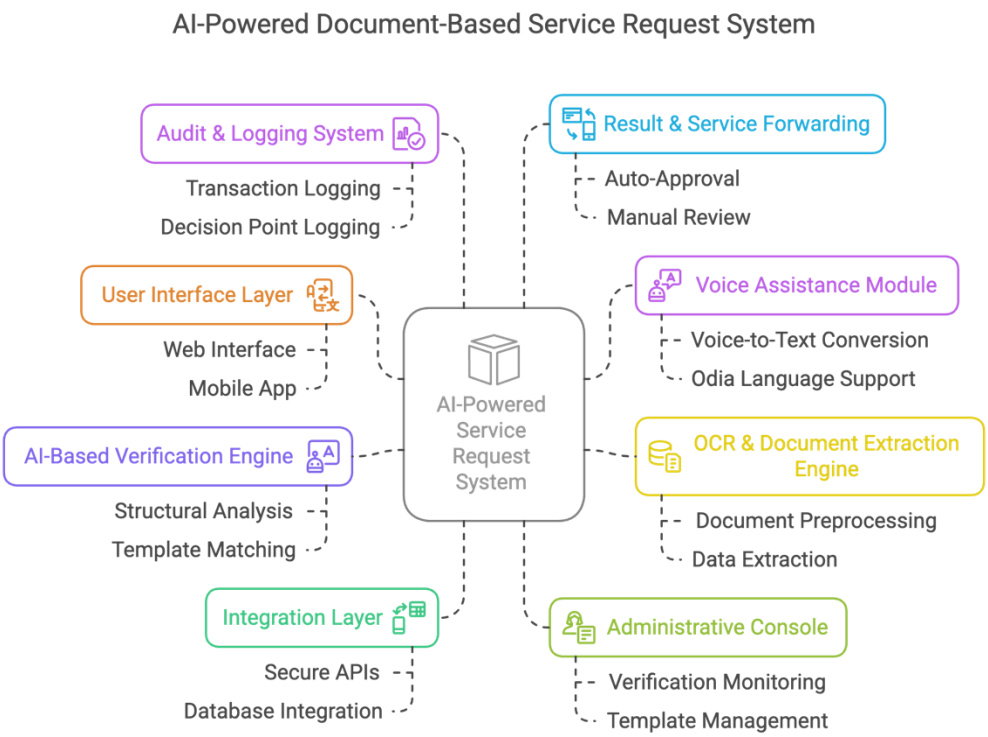
b. **Data Localization and Sovereignty Compliance:** No citizen data (including documents, personal identifiers, extracted fields, or any intermediate outputs) shall be transferred, transmitted, processed, or stored on servers located outside India at any stage of the solution lifecycle. All API calls, data flows, storage, and processing must be confined to government-approved, India-based hosting environments.

2. **Hardware Specification and Resource Planning:** The bidder shall propose the required hardware specifications (such as GPU servers, high-performance computing nodes, storage, network devices, etc.) necessary for optimal functioning of the AI models. The proposal must include sizing calculations based on anticipated transaction volumes, concurrency requirements, model complexity, and future scalability. The hardware should be proposed in a cost-effective and optimized manner, avoiding over-provisioning, while ensuring high performance, reliability, and security.
3. **Solution Architecture and Deployment Approach:** The bidder must provide a detailed solution architecture depicting the on-premises deployment model, including resource allocation, network topology, data flow diagrams, security zones, and failover mechanisms. The architecture should ensure high availability, disaster recovery, secure API gateway management, and compliance with State and National cybersecurity guidelines.
4. **No External API or SaaS Dependency:** All AI/ML models must operate within the Government's controlled infrastructure without dependency on external APIs, SaaS (Software as a Service) offerings, or cloud-based AI inference/processing engines that are outside the scope of the on-premises setup.
5. **API Development and Integration:** The System Integrator shall develop secure APIs to connect the AI-based Document Verification Layer with the SPDP core platform, citizen service portals, and issuing authority APIs. All APIs must operate within Indian infrastructure, using secure authentication (e.g., OAuth 2.0) and encryption standards. APIs should enable document submission, verification result retrieval, and seamless data exchange, supported by comprehensive API documentation, sandbox environments, and technical assistance for integrating applications.
6. **Security and Audit Readiness:** The deployed environment must conform to the security guidelines issued by NIC, MeitY, CERT-In, and any other applicable

government bodies. The solution must be auditable, with full traceability of data processing activities and compliance with security standards and periodic vulnerability assessments.

7. **Specify Hardware Scalability**: Provide clearer guidelines on hardware scalability, such as supporting 100,000 daily transactions initially with the ability to scale to 1 million within 2 years. For instance, "The hardware must support 100,000 daily verifications with <5-second latency and scale to 1 million transactions by 2027."
8. **Open-Source Preference**: Encourage the use of open-source AI frameworks (e.g., TensorFlow, PyTorch) to reduce costs and ensure flexibility. For example, "The SI shall prioritize open-source AI/ML frameworks to minimize licensing costs and ensure long-term maintainability."

10.7 Component Overview and Process Flow



10.8 Implementation Strategy

The implementation will follow an agile, phased approach ensuring timely delivery, feedback incorporation, and successful adoption across the state. Each phase builds incrementally on the previous one.

Phase 1 – Planning and Design

- Identify and finalize the list of government services and associated document types to be supported in the initial rollout.
- Define detailed workflows for each service, including submission, verification, escalation, and approval paths.
- Design intuitive, citizen-friendly UI/UX for the mobile application and web portal, ensuring accessibility and localization.

Phase 2 – AI Model Training

- Collect diverse real-world samples of each document type (across years and formats) to reflect variations in layout, quality, and language.
- Annotate data to identify key fields, validation logic, and visual markers (e.g., seals, QR codes).
- Train OCR and verification models using supervised ML techniques with performance benchmarking (target accuracy thresholds ≥ 90%).
- Establish continuous model evaluation and feedback loops for self-learning capability.

Phase 3 – Development

- Develop the AI-based document verification engine, with modular support for OCR, template matching, and authenticity scoring.
- Build API integration layer for secure data exchange with issuing authorities and service portals.
- Develop the admin web portal with dashboards, access control, audit logs, and configuration tools.
- Integrate offline-first capabilities and speech-to-text libraries.

Phase 4 – Testing

- Conduct comprehensive functional and integration testing across modules.
- Perform load testing to validate performance under expected concurrency levels.

- Ensure end-to-end security audit checks and data protection compliance.
- Organize UAT (User Acceptance Testing) with a selected group of citizens and government users to validate usability and process flow.

Phase 5 – Integration & Pilot Launch

- Integrate with selected backend systems such as e-District, SPDP, or other mission-mode applications.
- Deploy a pilot in 1–2 districts representing varied demographics and infrastructure levels within 3(?) months
- Monitor real-time performance, collect feedback, and refine the system based on practical learnings.

Phase 6 – State-wide Rollout

- Incorporate feedback and optimize model performance and UI based on pilot feedback.
- Expand deployment across all identified services and districts in a phased manner.
- Enable continuous support, training, and helpdesk mechanisms for officials and citizens.
- Establish monitoring dashboards for real-time analytics and future AI model retraining.

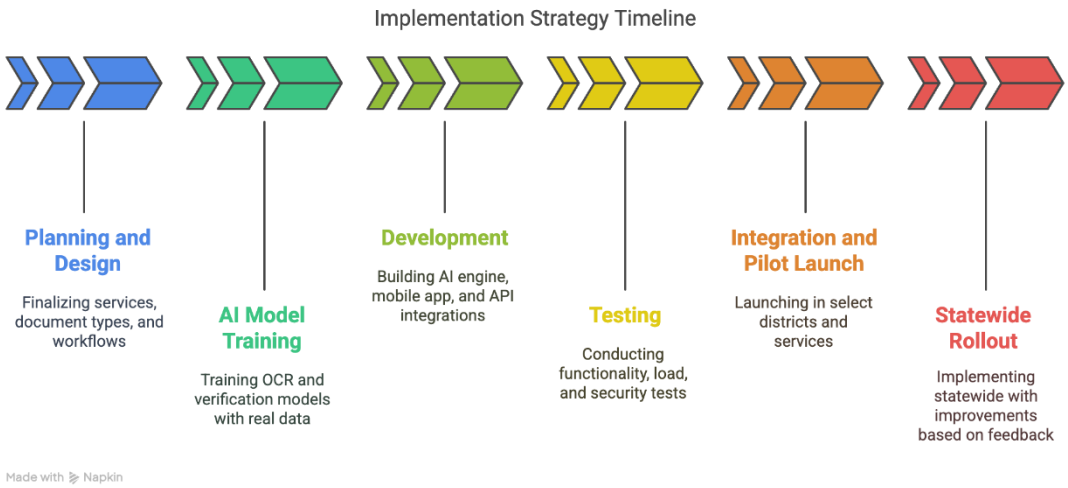


Fig 3.3 Implementation Strategy

10.8.1 Integration with Existing Systems

To ensure seamless service delivery and eliminate redundant document submissions, the solution must integrate with a range of government systems. These integrations are essential to verify authenticity, retrieve citizen-related data, and auto-populate forms for efficient processing.

The system should support both push and pull of data to and from systems proposed to be integrated. It is required that a standard mechanism of data exchange should be built and implemented using an industry specified data exchange protocol through a secure channel. The SI will have to co-ordinate with OCAC will facilitate this process. In addition, the solution should be designed in such a way that any future integration does not require any changes to the system.

The following systems are identified as mandatory for integration:

1. **SPDP (Social Protection Delivery Platform):** To retrieve demographic and entitlement information from the master citizen database for real-time validation.
2. **e-District:** For synchronization with certificates and service applications submitted by citizens, and to avoid duplication of efforts.
3. **SAMS (Student Academic Management System):** For academic record verification such as mark sheets and admission details during scholarship or education service requests.
4. **e-Shramik:** To validate worker registration details, generate or update digital cards, and auto-fetch employment history.
5. **Unorganised Workers Application:** For pulling worker profiles, benefit histories, and ensuring scheme eligibility for applicants from unorganised sectors.
6. **Krushak Odisha:** To validate farmer registration, landholding data, and subsidy or financial assistance eligibility through cross-verification.
7. **Odisha State Scholarship Portal:** For verifying scholarship records, sanction status, and validating previous disbursement history.
8. **Go Sugam:** Integration for streamlined access to agricultural and allied sector schemes and services.

9. **Ration Card Management System:** For identity verification, household validation, and address proof purposes.
10. **Schemes of Textile & Handloom Department:** To cross-reference beneficiaries, validate artisan identities, and prevent duplicate claims.
11. **CM KISAN:** To validate disbursal records, land ownership linkage, and streamline repeat applications.
12. **Subhadra Yojana:** For cross-checking female beneficiary details and ensuring rightful eligibility for newly introduced schemes.
13. **Birth and Death Registration System (OBDRS):** To auto-fetch proof of birth/death for use in family pension, widow schemes, or educational admission scenarios.
14. **Future Integration Flexibility**: Require a modular API design to accommodate future systems without significant rework. For example, "APIs shall follow a microservices architecture to enable easy integration with new systems within 3 weeks."
15. **Real-Time Data Sync**: Mandate real-time data synchronization to avoid discrepancies. For example, "The system shall ensure real-time data sync with SPDP and e-District, with a maximum latency of 2 second for critical data."

10.8.2 Prerequisites

The SI to follow and ensure following prerequisites before the requirement study

a) Consultation meeting with OCAC team
b) Identify and engage subject matter expert(s) as per the need
c) Readiness with the industry standard template for FRS and SRS documents
d) Readiness with the project traceability matrix template

10.8.3 Understand the Applications / Databases

The SI will perform all the functions and services necessary to accomplish the transition of the entire knowledgebase, application, infrastructure and services

under existing applications / databases from the present SI(s). The SI will be responsible for the overall management of the transition in accordance with the transition plan and will work to ensure the transition is completed on schedule and to identify and resolve any problems encountered. Further the SI will demonstrate its understanding of existing applications / databases and ability to support to reasonable satisfaction of OCAC, prior to the completion of Transition Phase, proving that it is ready to take over independently. The responsibility of the SI during the phase includes following.

a) Submit a detailed Knowledge Transfer plan, listing all the activities from their end, including the expectations from the OCAC.

b) Preparation of a checklist to ensure proper knowledge transfer which will be reviewed and approved by OCAC.

c) OCAC will facilitate the training on existing applications / databases, operations manuals, procedure manuals, and source code control and deployment / installation guide.

d) Hands-on exposure to existing applications / databases would be facilitated by OCAC.

e) The SI shall assess and validate the existing IT assets and inventories related to this project and submit its report in the gap assessment report as per the project timeline.

10.9 Integration Methodology – Development of Common APIs

10.9.1 The bidder shall be responsible for designing and developing standardized, reusable APIs to enable seamless integration with multiple departmental systems. These APIs must be published and made consumable via the Odisha Data Exchange Layer of the SPDP platform, ensuring secure and uniform access for all departmental applications. The cost quoted by the bidder must include the complete development and deployment of these common integration APIs, and no additional payment will be made for integrations performed with individual departmental applications using the standard API framework.

10.9.2 In scenarios where specialized or non-standard integration requirements arise such as unique data transformation logic, proprietary protocol support, or one-off connectors that fall outside the common API specifications, the effort will be treated as an exceptional integration case. In such cases, the payment will be made based on the mutually agreed actual integration cost, subject to departmental approval.

10.10 Testing

a) Provide the testing strategy including Traceability Matrix, Test Cases and Conduct Testing of various components of the software developed / customized as per industry standards for Software Testing Life Cycle.

b) Details of the testing strategy and approach should be provided in the response.

c) Identify, inform regarding testing requirements along with its impacts and work in a manner to satisfy all the testing requirements by adhering to the testing strategy outlined.

d) Ensure deployment of necessary resources and tools during the testing phases and perform solution testing based on the approved test plan, document the results and fix the bugs found during the testing.

e) Make sure that the end product delivered meets all the requirements specified in the document.

f) Take remedial action based on outcome of the tests.

g) Provide complete support to the departmental officials or their representatives at the time of User Acceptance Testing (UAT).

h) Ensure that all issues raised during UAT are closed and signed-off from respective authority.

i) Ensure that each module & features developed under this RFP is tested as per the latest version of the IEEE 730 (Software Quality Assurance Processes) standards and comply with GIGW guideline.

10.11 Cyber Security Audit

a) The SI shall ensure that the solution is in compliance with the CERT-In Security Policy and Guidelines.

b) The SI shall appoint CERT-In empanelled auditor who shall be responsible for performing the security audit of the solution.

c) The cost of audit & rectification of non-compliances shall be borne by the SI

d) Carryout security audit before go-live of application and obtain the safe-to-host certification.

e) Conduct periodic audit & certification as and when it is required as per the OSDC/Cloud policy.

f) The audit shall be performed at least on the below mentioned aspects.

   i) Functional Testing

   ii) Accessibility Testing

   iii) Application Security Audit

   iv) Vulnerability Testing

g) The illustrative deliverables for this activity are mentioned below

Activity	***Responsibility***
First Round Audit Report	Auditor
Rectified solution and submission of next round of audit	SI
Next Round Audit Report	Auditor
If required, rectified solution & submission of next round of audit	SI
Compliance Confirmation	Auditor

10.12 SSL Certification

a) Secure connection between client and server through Secure protocol HTTPS

b) Encryption of Data during transmission from server to browser and vice versa

c) Encryption key assigned to it by Certification Authority (CA) in form of a Certificate.

d) SSL Security in the application server.

10.13 Deployment & Configuration

The deployment of the solution will be fully hosted on-premises within the Odisha State Data Centre (SDC), ensuring strict adherence to data sovereignty and compliance requirements. High-performance GPU-enabled servers will be used to handle the AI processing and document verification tasks efficiently. The system will be designed for high availability, incorporating disaster recovery, failover mechanisms, and scalable architecture to ensure uninterrupted service. Security will be a top priority, with a NIC-certified security stack, audit trails, and regular vulnerability assessments to maintain compliance with government standards.

- **On-Premises Hosting at SDC using GPU-Enabled Infrastructure:**

  The entire solution will be deployed within the Odisha State Data Centre (SDC) to ensure compliance with state IT policies and enable better administrative control.

- **High Availability & Failover:**

  The architecture will include clustered nodes with automated failover mechanisms and periodic backups. DR (Disaster Recovery) setup will be maintained either within a secondary SDC zone or a separate secure facility.

- **Load Balancing & Scalability:**

  A horizontal scaling strategy will be implemented using containerized services behind load balancers (e.g., HAProxy/Nginx) to manage peak traffic, especially during scheme deadlines or state-wide campaigns.

- **Security & Compliance:**

  Deployment will adhere to NIC guidelines with hardened OS configurations, firewall layers, vulnerability assessment protocols, and centralized audit logging. Periodic third-party security audits will be conducted.

The System Integrator (SI) will be responsible for deploying new features and changes to the existing application using the hardware infrastructure provided by the Odisha State Data Centre (OSDC). Key responsibilities include:

a) **End-to-End Management:** The SI is accountable for the complete management of hosting and deployment of the application, which is already deployed on the OSDC's hardware infrastructure.

b) **Configuration, Installation, and Hosting:** The SI will configure, install, and host the application in a High Availability (HA) mode at OSDC, ensuring that

the system remains accessible and operational even during failures or maintenance.

c) **Disaster Recovery (DR) Compliance:** The SI must ensure that the deployment process complies with the Disaster Recovery (DR) policy set by OSDC, ensuring system resilience and business continuity.

10.14 UAT and Go-Live

After the completion of the development work for the application, the OCAC will conduct technical reviews of the development work performed by the System Integrator (SI) as part of the User Acceptance Testing (UAT). The SI will be responsible for:

a) **Preparation and Submission of Test Strategy and Results:** The SI must prepare and submit the test strategy, test cases, and test results for review. This will include a comprehensive breakdown of the testing approach and the outcomes of the tests conducted.

b) **Demonstration of Module-Wise Functionalities/Features:** The SI is required to demonstrate the functionalities and features of the application, module by module, in a staging environment to the DA&FE team.

c) **Support for Testing:** The SI must support DA&FE and its designated authority in conducting the testing. This includes providing necessary access to the systems as required by DA&FE for testing and review purposes.

d) **Rectification of Issues:** The SI is responsible for addressing any issues, bugs, improvements, enhancements, or up-gradations suggested by the DA&FE departments during the UAT phase. The SI must resolve these issues without any additional cost.

e) **Incorporation of Suggestions and Hosting in Production:** After incorporating all the suggestions made during the UAT phase, the SI must host the application in the production environment. Once the application has achieved a certain level of transaction and has undergone successful UAT, the go-live phase will be declared.

10.15 Post Implementation Support

10.15.1 Application Maintenance

a) Fixing of bugs & error as and when required in the existing & new modules developed

b) The defects will be covered which occurs due to development error(s)

c) Minor changes to the business process will be addressed except new table, database, etc.

d) Maintaining the updated version of source code

e) Tuning of the system to improve performance

f) Quality audit compliance (if applicable)

g) Enhancement of MIS report as per the requirement

h) Database query report management on emergency

i) Optimization of the already developed reports

j) Tuning of transactions

k) User & access management

10.15.2 System Support

a) Provide integration and user support on all supported servers, data storage systems

b) Management & monitoring of server infrastructure in cloud & Odisha state data centre

c) Installation and re-installation of the database

d) Application load balancing and database clustering

e) Network configuration

f) Perform Database, event & system log analysis

g) Database log management

h) Database & file back-up

i) Patch update

10.15.3 Operation Support

a) Provide support for user registration and access management

b) Provide guidance to resolve any wrong entry

c) First level analysis of the query and provide required clarification

d) Communicate the query to the application support team for resolution, if unable to provide required support for query resolution

e) Provide required reports to Government/department for smooth Implementation of the program

f) The bidder has to deploy a dedicated support resource for coordination with different stakeholders. The qualification and experience of support is as follows.

Parameter	Requirement
Education & Experience	– Graduate from any recognized university/institution. – Must have knowledge in the system and AI related processes – Must have proficiency in Odia language – Minimum 2 years of experience
Key Responsibilities	– Coordination with different stakeholders as well as development and O&M Team

(OCAC reserves the right to increase number of support resources in case of requirement)

g) If any technical resources are required during the operation and support phase, the bidder shall provide them at the man-month rate quoted under SL# 1.9 'Software Enhancement Service' in the price bid.

10.16 Guiding Principles

The solution should adhere to the following principles.

10.16.1 Standards

a) The system architecture should be based on industry standards and protocols

b) The system shall be centrally deployed and globally accessed

c) The system shall be designed to be scalable and easily extensible

d) The system should be flexible to cater to changing business, industry and compliance requirements (including reporting requirements in proper formats)

10.16.2 Application

a) All applications must take into account appropriate security, performance, efficiency and maintainability issues.

b) The ownership of the product licenses would be with OCAC

c) Upgrade to new releases should not become mandatory for the next five years from the date of installation.

10.16.3 Integration

The integrated solution design should include framework for integration of both internal and external applications and services using suitable architecture.

10.16.4 Data

a) Data will be owned, shared, controlled and protected as a corporate asset of the OCAC.

b) Data should only be accessed through application / interfaces to create, update and delete. There should not be any direct access to the data layer for users.

10.16.5 Data Security

a) Provide strategy to maintain data security at the application level, database level, messaging and middleware level

b) Provide security strategies when the applications are accessed by the resources from outside the network

c) Provide strategies of encryption and security for external transaction with partner network and systems

10.16.6 Adherence to Standards

The system shall comply with relevant defined industry standards (their latest versions as on date) wherever applicable. This shall apply to all the aspects of solution including its design, development, security, installation, and testing. The suggested architecture must be scalable and flexible for modular expansion. It should ensure ease of integration with software / applications developed using

common industry standards, since the solution may be linked and connected to other sources (websites, contents, portals, mobile app systems of other user departments etc.) as well as there may be loose/tight integration with backend system of other departments depending on individual service processes. The solution architecture should thus have provision to cater to the evolving requirements of the Department.

A reference list of the minimum industry standards which the system components should adhere to is mentioned below:

Component	***Standards***
Information Access / Transfer Protocols	SOAP, HTTP/HTTPS
Interoperability	Web Services, Open Standards
Portal Development	W3C Specifications
Document encryption	PKCS specification
Information Security	ISO 27001 certified System
Operation	ISO 9001 Certified
Service Management	ISO 20000 specifications or latest
Project Documentation	IEEE/ISO Specifications for documentation
Data Standards	All-important data entities should be in Line with standards published by DeiTY.

10.17 Security, Integrity & Confidentiality

a) **Web Services Security:** System shall comply with all the Web services including routing, management, publication, and discovery should be carried out in a secure manner. Those who are using the Web services should be able to utilize security services such as authentication, authorization, encryption and auditing. Encryption of data shall take place at client level itself. Application server shall provide SSL security.

b) **Data Integrity and Confidentiality**: Data integrity techniques need to be deployed to ensure that information has not been altered, or modified during transmission without detection. Similarly, Data confidentiality features are also to be applied to ensure that the data is only accessible by the intended

parties.

c) **Transactions and Communications:** With respect to the Data Transactions and Communications, system needs to ensure that the business process are done properly and the flow of operations are executed in correct manner.

d) **Non Repudiation Security:** The application shall have the Non-repudiation security services to protect a party to a transaction against false denial of the occurrence of that transaction by another party. End-to-End Integrity and Confidentiality of Messages, integrity and confidentiality of messages must be ensured even in the presence of intermediaries.

e) **Database Controls:** The database controls for online transaction processing systems like access to database directly, access to database through application, access to log files, access by the remote terminals, DBA controls, backup policy and backup procedures.

10.18 General Terms

10.18.1 Change Request Management

Any requirement beyond the scope mentioned in the RFP will be treated as Change Request and the process to address the change request is as follows:

- Identification and documentation of change request requirement– The details of scope of change will be analysed and documented
- Effort Estimate – OCAC will ask the successful bidder to submit the effort estimate in terms of man month rate using any standard estimation model.
- Approval or disapproval of the change request – Technical Committee constituted by OCAC will approve or disapprove the change requested including the additional payments, after analysis and discussion with the bidder on the impact of the change on schedule.
- Implementation of the change Request– The change will be implemented in accordance with the agreed cost, effort, and schedule. The vendor shall consolidate all approved CRs and raise invoice to OCAC accordingly.
- Additional cost towards change request management is to be agreed upon mutually by bidder and OCAC and factored separately post approval of efforts by OCAC. This clause should only be invoked for

justified business cases during the tenure of the project for such cases which is beyond the scope of this RFP.

The costing of change request shall be finalized as per cost mentioned in financial bid format- Software Enhancement Service. The bidder have to quote for 20 man month and payment shall be made as per actual consumed.

10.18.2 Exit Plan

a) Provide systematic exit plan and conduct proper knowledge transfer process to handover operations to OCAC technical team at least three months before project closure.
b) OCAC will work closely with the SI during knowledge transfer of testing, staging and production environment.
c) All knowledge transfer should be documented and possibly recorded.
d) Ensure capacity building of the IT resource persons of OCAC on maintenance of software and infrastructure.

10.18.3 Project Documentation

Below list of documents needs to be submitted to OCAC during the project contract period, as per the requirement of OCAC.

a) Latest version of Source Code
b) System Requirement Study Documents
c) System Design Document
d) Test Plans and Reports
e) Issue Logs
f) User Manual
g) Application Installation & Configuration Manual
h) Report of Security Audit & Safe-to-Host Certificate
i) Any other documents defined under Timeline & Tentative Deliverables

All the above documentation should be done as per IEEE/ISO/CMM Standard.

10.19 Time Line & Tentative Deliverables

T: Date of issuance of Purchase Order or signing of contract whichever is earlier

Sl#	*Project Component*	*Tentative Deliverables*	*Responsibility*	*Time line*
a)	Mobilization of Team and System Study	– Final Project Schedule with breakdown structure – Detailed Team Structure with team members – Point of Contact – FSR/SRS Document with screen prototypes and Prototype walk through	Bidder	T+2 Weeks
b)	Approval of SRS	– Approval letter	OCAC	T+3 Weeks
c)	Software Development, Testing, Deployment, Configuration	– Source Code – System Design Document – Test Plans & Test Cases – Operation manual – Configuration Manual – Administration Manual – Hardening checklist (if any) – Security Policy document – FAQs – Trouble Shooting Guide/ Handbook for helpdesk – Hosting of Application in staging environment – Load Testing report – Performance tuning parameters for fine tuning application on server	Bidder	T+ 9 Weeks
d)	User Acceptance Test	– Preparation Test Cases – Conduct of UAT	Bidder & OCAC	T+10 Weeks
e)	Training	– Training to Stakeholders	Bidder & OCAC	T4+11 Weeks

Sl#	*Project Component*	*Tentative Deliverables*	*Responsibility*	*Time line*
f)	Security Audit	– Auditor's vulnerability report – Fixing of vulnerabilities found during security audit – Safe to Host to be issued by auditor	Bidder	T+13 Weeks
g)	Go-Live	– Movement of application from Staging to Production environment	Bidder	T+14 Weeks
h)	Post Implementation Support (operation & management)	– Quarterly status report on application wise document verification made – Monthly log of service calls received from various applications and problem resolution reports – Quarterly Performance Monitoring Reports for the system – Updated system design documents, specifications for every change request, if any – Latest source code, application deployment files, configuration files for entire solution (to be submitted for 1st quarter and subsequently after each security audit/any change in application/on change request) – Updated user manuals, administration manuals, training manuals etc. in every change request – Security Audit report and safe to host certificate issued by Cert-in empanelled firm every year.	Bidder & OCAC	**3 Years from Go-Live**

1. **Define "Few Seconds" Precisely**: Specify exact latency targets, e.g., "Document verification shall complete within 5 seconds for 95% of cases under normal load (10,000 transactions/day)."
2. **Peak Load SLAs**: Include SLAs for peak loads, such as during scholarship or ration renewal seasons. For example, "During peak loads (50,000 transactions/day), the system shall maintain 99% uptime and 7-second verification latency."

10.20 Payment Terms

a) 40% of cost of Application will be paid after UAT of application

b) 20% will be paid after successfully running of application for a period of Three (3) month from the date of Go-live of application

c) The remaining 40% Cost of Application will be paid equally in 4 QGRs from the date of Go-live

d) Cost of Post Implementation Support (operation and management) will be paid equally in 12 QGRs (QGR Start will be made from the date of Go-live)

e) 100% cost of the security audit will be released after submission of auditors report and Safe-to-Host Certificate of respective audit.

f) 100% of the onsite resources will be paid on quarterly basis.

g) Infrastructure Components– 90% payment after delivery of infrastructure/hardware component and balance 10% after successful commission of hardware environment.

h) System Software – 100% cost of the system software (like Operating system, application server, database server etc. shall be paid after supply of required licenses/subscriptions

i) Secure Socket Layer (SSL) – 100% of the SSL cost shall be paid after installation of certificates

j) Change Request – 100% cost of the respective change request cost shall be paid after UAT

k) Integration - – 100% cost of the respective integration cost shall be paid after UAT

l) On-boarding of new type of document – 100% cost shall be paid after 3 months from the date of UAT.

m) Taxes will be paid extra as per the rate prevalent at the time of billing

n) Payment shall be made within 30 working days of the receipt of invoice along with supporting documents subject to penalties, if any.

10.21 Performance Requirement (SLAs)

The purpose of this Service Level Agreement (herein after referred to as SLA) to clearly define the performance criteria that shall be adhered by the SI during the contract period.

Sl#	Major Area	Parameter	Requirements	Penalty
a)	Availability of application	Application covering all the features	98% availability round the clock and Computation will be done on monthly basis. Note: Fault at application level only	Up to 90-97.99% - 1% of application development cost. Less than 90%- 2% of application development cost
b)	Resolution Time for the Application (Bug fixing)	Time taken by the Bidder to fix the problem	Within 6 hours of reporting	6hrs to 24 hrs @0.01% of application development cost. Beyond 24 hrs 0.1% of application development cost.
c)	Resolution Time for Authentication / Registration related issues	Time taken by the Bidder to fix the problem	Within 3 hours of reporting	3hrs to 6 hrs @0.01% of application development cost. Beyond 6 hrs 0.1% of application development cost.

Note: Penalty will be imposed on operation and maintenance support cost of respective QGR. Maximum ceiling of the penalty will be 10% cost of the operation and maintenance support of respective year.

10.22 Waiver of Penalty

If at any time during the contract, the selected SI should encounter conditions impending timely performance of service, the selected SI shall promptly notify to OCAC in writing of the fact of the delay and its likely duration along its cause(s). As soon as practicable after receipt of the selected SI's notice, OCAC shall evaluate the situation and may at its discretion waive the penalty on the request of the selected SI.

10.23 AI related infrastructure

10.23.1 Indicative Bill of Quantity (BOQ)

Sl. No.	Item Description	Quantity
01	Edge AI Runtime Environment	3 no's
02	Core Hosting Node	3 no's
03	Virtualization Software (for above all the hardware covered under Edge AI Runtime Environment and Core Hosting Nodes)	1 lot.

(This is a minimum indicative BoQ, the bidder may add more item/quantity as per their proposed solution)

The bidder must furnish the MAF from OEM against all the hardware and System Software Items quoted.

10.23.2 Minimum Technical Specifications of Edge AI Runtime Environment

Edge AI Runtime Environment				
Sl.	**Parameter**	**Minimum Specification / Compliance required**	**Compliance (Yes/No)**	**Document Reference**
1.	Make & Model:			
2.	Form factor	Maximum 2U rack mount server with Bezel, Bezel Locking Kit, Chassis Intrusion Detection Kit, Rail kit, and AC power cords.		
3.	Processor	Minimum 2 x Latest 5th Generation or higher 2.1GHz, 32-core minimum, 160MB or higher L3 cache, 3 UPI or higher. 64-it		

		x86 processor fully binary compatible to 64/32-bit applications and supporting hyper-threading. Number of cores on a single die/socket will be treated as a single processor.		
4.	Memory	768 GB RDIMM, using 64GB or higher DDR5 modules in balanced configuration, scalable up to 2TB. Minimum 4800 MT/s. Advanced ECC to protect servers against single-bit errors as well as to protect against multi-bit memory errors within a single RAM chip as well as within a single memory module. Advanced Error-correcting code (ECC), online spare, mirroring, and Fast Fault Tolerance Adaptive Double DRAM Device Correction (ADDDC).		
5.	Storage controller	Tri-mode SAS/SATA/NVMe RAID controller with minimum 8GB or higher cache, & RAID 1/5/6/10/50/60 support. Offered controller must support mix-and-match up to 16 no's 12G SAS, 6G SATA, and 16G NVMe drives to the same controller.		
6. 6	Hard Disk Drive	2 X 960GB RI NVMe SSD.		
7.	Video Controller	Video modes up to 1920 x 1200@60Hz (32 bpp).		
8.	GPU	4 x NVIDIA L40S/48GB GPU accelerators in air-cooled system configuration.		
9.	Network ports	Minimum 4 x 1G (RJ-45), 2 x 2-port 25G(SR) Ethernet, 2 x 2-port 32G(SW) FC.		
10.	OS & Hypervisor certification	Certified for latest version of Red Hat Enterprise Linux, SUSE Linux Enterprise Server, Ubuntu, Microsoft Windows Server, HVM, & VMWare.		
11.	Power Supply	Minimum Hot Plug Redundant power supplies of maximum 2.2kW or better.		
12.	Fans/Blower	Fully populated redundant (N+1) hot-swap fans system		

13.	Other Interfaces	Minimum 1 x 1Gbps Dedicated OOB system management port (RJ-45), 1 x video port, 3 x exterior USB 3.0 ports		
14.	System Compliances	ACPI 6.3, PCIe 5.0, SMBIOS 3.2, UEFI 2.7, IPMI 2.0, AES/3DES, SNMP v3, TLS 1.2, RESTful API, ASHRAE A3/A4		
		Continuous, proactive health monitoring as well as notification of actual or impending component failure alerts on key internal server components such as CPUs, memory, temperature, fans, RAID controllers, hard drives (including cache modules) and power supplies.		
15.	System BIOS	System should boot with & run the BIOS from the same server hardware OEM (manufacturer). All updates should happen only using quoted OEM's access controller & management software to enforce security.		
16.	System Management & Security	Silicon-based Hardware Root of Trust		
		FIPS 140-2, CNSA, TPM 2.0		
		UEFI SECURE BOOT or Cryptographically verified trusted booting meeting NIST SP 800 standards		
		Secure erase		
		SECURE RECOVERY or Continual runtime detection, alerting, and with ability to rollback/recover BIOS & firmware		
		Should help to proactively identify out-of-date BIOS, drivers, and Server Management agents and enable the remote update of system software/firmware components. Should have dashboard for firmware baselines while performing minimum required firmware checks and highlighting out-of-compliance devices for updates with the selected firmware baseline.		
17.	System Management	System management software shall be from the same server hardware OEM and shall be with perpetual license.		

		System remote management should support browser based graphical remote console along with Virtual Power button, remote boot using any of USB/CD/DVD Drive. It should be capable of offering upgrade of software and patches from a remote client using Media/image/folder; It should support server power capping and historical reporting and should have support for multifactor authentication		
		Server should support monitoring and recording changes in the server hardware and system configuration. It assists in diagnosing problems and delivering rapid resolution when system failures occur		
18.	Serviceability	System should support embedded remote support to transmit hardware events directly to OEM or an authorized partner for automated phone home support		
19.	Certifications & benchmark	Relevant product datasheets, certification, benchmark or reference documents verifiable from publicly available data sources must be submitted along with bill of material for detailed bid evaluation. Links for certifications to be submitted.		
20.	Warranty	Five years on-site comprehensive OEM Warranty Support with 24X7 coverage and access to OEM TAC/support. OEM shall have their own support portal to log the case online and historical data about cases must be available in the same portal.		
21.	IDC Ranking	OEM should be ranked within top 3 as per IDC report for any one of the previous four quarter in India for server.		

10.23.3 Minimum Technical Specifications of Core Hosting Node

Core Hosting Node				
Sl.	**Parameter**	**Minimum Specification / Compliance required**	**Compliance**	**Document**

			(Yes/No)	Reference
1.	Make & Model:			
2.	Form factor	Maximum 2U rack mount server with Bezel, Bezel Locking Kit, Chassis Intrusion Detection Kit, Rail kit, and AC power cords.		
3.	Processor	Minimum 2 x Latest 5th Generation or higher 2.1GHz, 32-core minimum, 160MB or higher L3 cache, 3 UPI or higher. 64-it x86 processor fully binary compatible to 64/32-bit applications and supporting hyper-threading. Number of cores on a single die/socket will be treated as a single processor.		
4.	Memory	768 GB RDIMM, using 64GB or higher DDR5 modules in balanced configuration, scalable up to 2TB. Minimum 4800 MT/s. Advanced ECC to protect servers against single-bit errors as well as to protect against multi-bit memory errors within a single RAM chip as well as within a single memory module. Advanced Error-correcting code (ECC), online spare, mirroring, and Fast Fault Tolerance Adaptive Double DRAM Device Correction (ADDDC).		
5.	Storage controller	Tri-mode SAS/SATA/NVMe RAID controller with minimum 8GB or higher cache, & RAID 1/5/6/10/50/60 support. Offered controller must support mix-and-match up to 16 no's 12G SAS, 6G SATA, and 16G NVMe drives to the same controller.		
6. 6	Hard Disk Drive	2 X 960GB RI NVMe SSD.		
7.	Video Controller	Video modes up to 1920 x 1200@60Hz (32 bpp).		
8.	Network ports	Minimum 4 x 1G (RJ-45), 2 x 2-port 25G(SR) Ethernet, 2 x 2-port 32G(SW) FC.		

9.	OS & Hypervisor certification	Certified for latest version of Red Hat Enterprise Linux, SUSE Linux Enterprise Server, Ubuntu, Microsoft Windows Server, HVM, & VMWare.		
10.	Power Supply	Minimum Hot Plug Redundant power supplies of maximum 2.2kW or better.		
11.	Fans/Blower	Fully populated redundant (N+1) hot-swap fans system		
12.	Other Interfaces	Minimum 1 x 1Gbps Dedicated OOB system management port (RJ-45), 1 x video port, 3 x exterior USB 3.0 ports		
13.	System Compliance s	ACPI 6.3, PCIe 5.0, SMBIOS 3.2, UEFI 2.7, IPMI 2.0, AES/3DES, SNMP v3, TLS 1.2, RESTful API, ASHRAE A3/A4		
		Continuous, proactive health monitoring as well as notification of actual or impending component failure alerts on key internal server components such as CPUs, memory, temperature, fans, RAID controllers, hard drives (including cache modules) and power supplies.		
14.	System BIOS	System should boot with & run the BIOS from the same server hardware OEM (manufacturer). All updates should happen only using quoted OEM's access controller & management software to enforce security.		
15.	System Manageme nt & Security	Silicon-based Hardware Root of Trust		
		FIPS 140-2, CNSA, TPM 2.0		
		UEFI SECURE BOOT or Cryptographically verified trusted booting meeting NIST SP 800 standards		
		Secure erase		
		SECURE RECOVERY or Continual runtime detection, alerting, and with ability to rollback/recover BIOS & firmware		
		Should help to proactively identify out-of-date BIOS, drivers, and Server Management agents and enable the remote update of system software/firmware components. Should		

		have dashboard for firmware baselines while performing minimum required firmware checks and highlighting out-of-compliance devices for updates with the selected firmware baseline.		
16.	System Manageme nt	System management software shall be from the same server hardware OEM and shall be with perpetual license.		
		System remote management should support browser based graphical remote console along with Virtual Power button, remote boot using any of USB/CD/DVD Drive. It should be capable of offering upgrade of software and patches from a remote client using Media/image/folder; It should support server power capping and historical reporting and should have support for multifactor authentication		
		Server should support monitoring and recording changes in the server hardware and system configuration. It assists in diagnosing problems and delivering rapid resolution when system failures occur		
17.	Serviceabilit y	System should support embedded remote support to transmit hardware events directly to OEM or an authorized partner for automated phone home support		
18.	Certification s & benchmark	Relevant product datasheets, certification, benchmark or reference documents verifiable from publicly available data sources must be submitted along with bill of material for detailed bid evaluation. Links for certifications to be submitted.		
19.	Warranty	Five years on-site comprehensive OEM Warranty Support with 24X7 coverage and access to OEM TAC/support. OEM shall have their own support portal to log the case online and historical data about cases must be available in the same portal.		

20.	IDC Ranking	OEM should be ranked within top 3 as per IDC report for any one of the previous four quarter in India for server.		

10.23.4 Technical Specifications of Cloud Management

Virtualization Software with Cloud Management				
Sl.	**Parameter**	**Minimum Specification / Compliance required**	**Compliance (Yes/No)**	**Document Reference**
1.	Make & Model:			
2.	The bidder shall propose direct back-to-back OEM support for the proposed solution with an unlimited number of incidents on a 24 x 7 x 365 coverage basis.			
3.	The bidder shall propose - Plan, Design and Implementation Services (Professional Services) from the Virtualization software OEM. Bidder shall ensure that offered virtualization software management are configured in N+1 high availability (HA). The virtualization software OEM shall not subcontract the design and implementation services to any third party. The OEM engineers designing, deploying and implementing shall have to be on the payroll of the virtualization software OEM.			
4.	Bidder should propose Virtualization software with Container runtime and cloud management from a single OEM for interoperability as well as seamless single-window TAC support experience for the end-customer. Bidder shall submit an undertaking from their proposed virtualization software OEM for overall ownership of installation, commissioning, integration and support during asked support duration of 5-years post-FAT.			
5.	Virtualization software solution shall be qualified for both Intel as well as AMD CPU architectures along with capability to provide high availability.			
6.	Virtualization shall support all leading Guest Operating Systems.			
7.	Virtualization shall support migration of a running virtual machine from one host to another within the same cluster with zero downtime.			

8.	Virtualization shall automatically restart virtual machines on another host in the same cluster in the event of an unexpected host failure within the cluster.		
9.	Virtualization shall dynamically schedule the placement of virtual machines within a cluster based upon optimal workload distribution across the cluster.		
10.	Virtualization Management software should be available independent of failure of host/node, and OS.		
11.	Virtualization shall support migration the virtual disk(s) of a running of virtual machine from one storage data store to another with zero downtime.		
12.	Virtualization shall support both expansion and shrinking of VM cluster		
13.	Virtualization shall support running virtual machines on external storage via iSCSI, NFS, and Fibre Channel without any dependency on CSI operator/driver.		
14.	Virtualization shall allow administrator to create service plan or t-shirt size based on CPU, Memory and Storage and shall be available to users while creating / provisioning the instance / VMs		
15.	Virtualization shall support codeless integration with OCAC's existing DNS/IPAM/DHCP solution from "Efficient iP" and also, shall support codeless integration with leading external providers like Infoblox, Microsoft DNS, BlueCat, SolarWinds, and phpIPAM etc.		
16.	The offered Hypervisor shall support below features for Virtual Machine Management: a) Create / Delete / Restart / Start / Stop / Suspend and Discovery of Virtual machines. b) Snapshot operations – Create / Delete / revert of Virtual machines c) Tagging – Add / Delete / Edit tagging for Virtual Machines. d) Live Migration of VM, VM HA and Pin virtual machine to a specific host. e) Cloning of VM, Clone to VM Template. f) Virtual Hardware Management - Add and remove virtual hardware such as hard disks, network interfaces, CPU and memory from a managed virtual machine.		

17.	Access to grouping of resources shall be controlled through appropriate roles while assigning to users. Roles shall provide access to resources using appropriate permissions. At least, it shall be possible to configure following key permissions: a) Access to Native data protection configuration. b) Read or full access for creation of Service plans. c) Access to API to executes scripts on Instances / VM. d) Access for allowing users to use Dynamic workload schedular for VM placement and pinning of VM to a specific host. e) Access for creating the automation scripts. f) Permission for resizing the instance. g) Access to instance / VM – Console, Adding or deleting an Instance / VM.		
18.	Virtualization solution shall include suitable multi-cloud management through additional cloud from a single-pane-of-glass common GUI without any additional coding.		
19.	Virtualization shall support as well as shall provide a common dashboard for various CNCF-certified Kubernetes distributions such as Docker, AKS, MKS, EKS, and GKE along with externally deployed OpenStack integration for cloud native application from day-1. Solution shall be provided to support multi-tenancy as well as autoscaling for such OpenStack deployments.		
20.	Virtualization should support both macro- and micro-segmentation from day-1.		
21.	Virtualization shall allow creation of Wiki, which shall be RBAC-controlled, auditable Wiki that allows easy access to information, notes, configurations or any other data needed to be referenced or shared with others.		
22.	Virtualization shall be capable of creating a cloud with secure multitenancy where each tenant shall have capability to have own uses, roles and integration with required authentication providers like Active directory, Okta, SAML, etc.		
23.	Virtualization shall be truly heterogenous and shall quickly integrate with below tools / Platforms /		

	integrations as well as shall provide consistent unified console for all below integrations for self-service provisioning and life-cycle management: a) Hypervisors: VMware, Nutanix, Microsoft, HVM etc. b) Clouds: AWS, Azure, GCP, Oracle, OpenStack etc. c) Identity: MS AD, Okta, SAML, LDAP, OneLogin etc. d) Network: NSX, ACI, Infoblox, Bluecat, SolarWinds etc. e) Backup: Veeam, Commvault, Zerto etc. f) K8's: AKS, EKS, GKE, Kubernetes, OpenShift, HVM cluster etc. g) Load Balancer: F5, A10, Citrix, ALB, Azure Load Balancer etc. h) Automation: Chef, Puppet, Ansible, Ansible Tower, Aria/vRA etc.		
24.	Virtualization shall quickly enable on-prem private clouds, centralize public cloud access, deploy Kubernetes clusters and execute automation jobs from the unified console instead of opening the separate console of above-mentioned integrations.		
25.	Virtualization shall have truly secure multi-tenant isolation environment where each tenant shall have capability for unique users, role and workloads.		
26.	Virtualization shall be truly-cloud agnostic so that all supported clouds can be provisioned from CMP console in a common repeatable and unified fashion.		
27.	Virtualization shall support Bare metal environment as Private cloud to deploy KVM, Docker, Kubernetes and other virtualization technologies and shall also support major platform like HPE OneView and CISCO UCS.		
28.	Virtualization shall have at least 50+ in-built templates or instance types across technology platform from different cloud providers of public cloud, private cloud, containerized and bare-metal environments.		
29.	Virtualization shall also provide the flexibility for creating the customized templates or instance types across technology platforms from different cloud providers of		

	public cloud, private cloud, containerized and bare-metal environments.		
30.	Virtualization shall also support a vast array of providers and configurations with programmatic markup or Infrastructure as Code capabilities i.e. Terraform, ARM (Azure), Kubernetes, CloudFormation (AWS) and Helm.		
31.	Virtualization shall provide common multi-cloud blueprints for programmatic configuration through Infrastructure-as-code for deploying across on-premises and public cloud.		
32.	Virtualization shall also allow create vmdk, qcow2, vhd and raw Images from scratch.		
33.	Virtualization should have suitable data backup solution included which shall be able to protect VM, Containers, Hosts, File, directory, Volume to Target Storage provider including backup of the management plane of virtualization.		
34.	Backup solution included with virtualization shall have deep integration into the instance provisioning window so that all newly created instances are protected and backed up automatically.		
35.	Virtualization shall support capabilities to integrating machine provisioning /management with configuration management databases (CMDBs), Ticketing systems, IP address management systems, or Domain Name System (DNS) servers.		
36.	Virtualization shall allow configuration of networks across all clouds and existing networks from Clouds added in the solution shall auto-populate in the Networks section. Virtualization shall support security groups which shall act as a virtual firewall and should support integration with multiple SDN solutions.		
37.	Virtualization shall provide comprehensive service catalog with capabilities for service design and lifecycle management, a web- based self-service portal for users to order and manage services.		

38.	Virtualization shall support creation of services such as 'Single VM' and a 'multi-tier application infrastructure' like service map across private and public cloud as catalogue within a group or project/applications.		
39.	Virtualization shall be able to give complete cost governance across the private, as well as public clouds.		
40.	Virtualization shall allow customers to create budgets across Private and public clouds to provide insights into spending. It shall be possible to assign the budget to account, tenant, user, groups and at tenant level.		
41.	Vendor shall provide the live costing API to pull down the overall costing details from the public cloud provider like AWS, Azure, google etc. and shall also have in-built cost metering for private cloud like VMware. It shall be possible to generate highly granular costing data through invoices for both public and private clouds.		
42.	Virtualization shall have in-built analytics engine to provide administrators the tools to break down costs and usage across group, cloud and at tenant level. Virtualization shall also show recommendations or guidance for resource and cost optimization by analyzing the CPU, RAM, and Storage activity.		
43.	Virtualization shall ensure that any guideline or recommendation shall not be executed automatically, and a decision shall be taken only after due diligence.		
44.	Virtualization shall facilitate the servicing of a large amount of Log data along with viewing and shall have in-built log engine such as Elasticsearch or equivalent. It shall be able to define the retention period for logs data from few days to years for PCI Compliance along with defining the syslog forwarding rules for exporting purpose.		
45.	Virtualization shall also provide the creation of Application monitor checks to measure the high availability of application provisioned either through blueprint or manually added. It shall be possible to define		

	the number of days to calculate the availability level of application along with check interval which shall be possible to define as low as 1 minute.		
46.	Virtualization shall also be able to generate an incident based on Application monitor checks and shall be able to integrate the same with well-known support ticketing platform.		
47.	Bidder shall provide the licenses for a total of 6 no's of 2-CPU servers, 32 physical-CPU-cores per-CPU. Offered licensing shall be scalable.		
48.	Offered licensing subscription shall be for at-least 5years from the date of FAT.		

10.24 General Conditions

a) Payment schedule - Payments to the SI after successful completion of the target milestones (including specified project deliverables), would be made as under: -

b) The supplier's/ selected SI's request for payment shall be made to the purchaser in writing, accompanied by invoices describing, as appropriate, the goods delivered and related services performed, and by the required documents submitted pursuant to general conditions of the contract and upon fulfilment of all the obligations stipulated in the Contract.

c) Due payments shall be made promptly by the purchaser, within thirty (30) days after submission of an invoice or request for payment by the supplier/ selected SI/authorized partner, and the purchaser has accepted it.

d) The currency or currencies in which payments shall be made to the supplier/ selected SI under this Contract shall be Indian Rupees (INR) only.

e) All remittance charges will be borne by the supplier/ selected SI/authorized partner.

f) In case of disputed items, the disputed amount shall be withheld and will be paid only after settlement of the dispute.

g) Payment in case of those goods which need testing shall be made only when such tests have been carried out, test results received conforming to the prescribed specification.

h) Any penalties/ liquidated damages, as applicable, for delay and non-performance, as mentioned in this bidding document, will be deducted from the payments for the respective milestones.

i) Taxes, as applicable, will be deducted/ paid, as per the prevalent rules and regulations at the time of billing. Legitimate payment shall be made within 30 working days of the receipt of invoice along with supporting documents subject to penalties, if any.

10.25 OCAC Responsibilities

a) Assign a nodal officer who will be single point of contact from the beginning of the project till successful implementation.

b) Assign the nodal officer/ Agency to handover the knowledge of the existing application

c) Provide necessary support to the development team of the SI for smooth execution of project.

d) Provide all the relevant documents and information during the system study and analysis.

e) Facilitate the Software Solution Provider for the third-party software integration.

f) Provide approval of SRS Document, User Acceptance Test certificate, Go-Live Certificate, approval of activity report during Operation & Maintenance Support phase, AMC etc.

g) Conduct exclusive hand-holding on the existing Application to the SI. Following activities shall be taken into consideration during handover process:

- Establish a transition team
- Create a transition plan
- Provide detailed documentation
- Ensure multifaceted knowledge transfer
- Get access to all third-party services
- Transfer codebase ownership
- Provide proper understanding on the source code, database table structure etc.

h) Provide hosting infrastructure along with SMS and Email Gateway etc.

11 Formats for Response

11.1 Pre-Qualification Bid Formats

11.1.1 FORM PQ-1: Cover Letter

(To be submitted on the Letterhead of Bidder)

To

The General Manager (Admin),
Odisha Computer Application Centre,
N-1/7-D, Acharya Vihar, P.O. RRL, Bhubaneswar - 751013.

Sub: Selection of Technology Partner For AI-Driven Document Verification Engine under Social Protection Delivery Platform

Madam/Sir,

I, the undersigned, offer to provide the services for the proposed assignment in respect to your <<RFP No: ……>>, <<date>>. We hereby submit our proposal which includes the pre-qualification proposal, technical proposal and commercial proposal, sealed under separate envelopes. Our proposal will be valid for acceptance up to **180 Days** and I confirm that this proposal will remain binding upon us and may be accepted by you at any time before this expiry date.

All the information and statements made in our proposal are true and correct and I accept that any misinterpretation contained in it may lead to disqualification of our proposal. If negotiations are held during the period of validity of the proposal, I undertake to negotiate on the basis of proposal submitted by us. Our proposal is binding upon us and subject to the modifications resulting from contract negotiations.

I have examined all the information as provided in your Request for Proposal (RFP) and offer to undertake the service described in accordance with the conditions and requirements of the selection process. I agree to bear all costs incurred by us in connection with the preparation and submission of this proposal and to bear any further pre-contract costs. In case, any provisions of this RFP/ ToR/Scope including of our technical and financial proposal are found to be deviated, then you shall have

rights to reject our proposal. I confirm that, I have the authority to submit the proposal and to clarify any details on its behalf.

I understand you are not bound to accept any proposal you receive.

Yours faithfully,
(Authorized Signatory)
Name, Designation & Contact No. and Seal

11.1.2 FORM PQ-2: Bidder's Organization (General Details)

(To be submitted on the Letterhead of Bidder)

Sl#	Information	Details
a)	Name of Bidder	
b)	Registered Address of Bidder	
c)	Address for Communication	
d)	Address of local office in Odisha. If bidder has no local office at the time of bid submission, an undertaking has to be furnished on bidder's letter head on setting up an office within 3 months from issuance of work order.	
e)	Name, Designation and Address of the contact person to whom all references shall be made regarding this RFP	
f)	Mobile no. of contact person:	
g)	E-mail address of contact person:	
h)	GST Number of the Firm	
i)	PAN No. of the firm	

Yours faithfully,

(Authorized Signatory)
Name, Designation & Contact No.
Seal

11.1.3 FORM PQ-3: Acceptance of Terms and Conditions

(To be submitted on the Letterhead of Bidder)

To

The General Manager (Admin),
Odisha Computer Application Centre,
N-1/7-D, Acharya Vihar P.O. RRL, Bhubaneswar - 751013.

Sub: Selection of Technology Partner For AI-Driven Document Verification Engine under Social Protection Delivery Platform

Madam/Sir,

I have carefully and thoroughly gone through the Terms & Conditions along with scope of work contained in the <<RFP No: ……>> regarding RFP for "RFP for Selection of Technology Partner for AI-Driven Document Verification Engine under Social Protection Delivery Platform".

I declare that all the provisions/clauses including scope of work of this RFP are acceptable to our company. I further certify that I am an authorized signatory of the company and I am, therefore, competent to make this declaration.

Yours faithfully,

(Authorized Signatory)
Name, Designation & Contact No.
Seal

11.1.4 FORM PQ-4: Project Citation Format

a)	Project Name:	
b)	Value of Contract/ Work Order (In INR):	
c)	Name of the Client:	
d)	Project Location:	
e)	Contact person of the client with address, phone and e-mail:	
f)	Project Duration:	
g)	Start Date (month/year): Completion Date (month/year):	
h)	Status of assignment: Completed / Ongoing (if it is on-going, level of completion)	
i)	Narrative description of the project with scope:	
j)	List of Services provided by your firm/company:	

11.1.5 FORM PQ-5: Manufacturer's Authorization Form (MAF)

Letter No.

To
The General Manager (Admin.)
Odisha Computer Application Centre
Plot No. - N-1/7-D, Acharya Vihar
P.O. - RRL, Bhubaneswar – 751013

Sub: OEM Authorization Letter RFP Ref No: OCAC-SEGP-INFRA-0002-2025-25052,
dtd.03/05/2025.

Dear Sir,
We, who are established and reputable _____having factories / development facilities at (address of factory / facility) do hereby authorize M/s_____(Name and address of Agent) to submit a Bid, and accept the Purchase Order against the above Bid Invitation.

We hereby extend our full guarantee and warranty for the Solution, Products and
services offered by the above firm against this Bid Invitation. We further undertake that we will support the supplier will all related spares and maintenance during the entire contract period.

We duly authorize the said firm to act on our behalf in fulfilling all installations, Technical support and maintenance obligations required by the Project.

Yours faithfully,
(Name)
Seal

Note: This letter of authority should be on the letterhead of the OEM and should be signed by a person competent and having the power of attorney to bind the manufacturer.

11.1.6 FORM PQ-6: Bank Guarantee Template

To
The General Manager (Admin)
Odisha Computer Application Centre
(Technical Directorate of E & IT Dept, Govt. of Odisha)
N-1/7-D, Acharya Vihar P.O. - RRL,
Bhubaneswar - 751013

Whereas <<Name of the bidder>> (hereinafter called 'the Bidder') has submitted the bid for Submission of <<RFP No: ……>> for Selection of Technology Partner for AI-Driven Document Verification Engine under Social Protection Delivery Platform (hereinafter called "the Bid") to OCAC.

Know all men by these presents that we <<Name of the Bidder>> having our office at <<Address>> (hereinafter called "the Bank") are bound unto the <<Nodal Agency>> (hereinafter called "the Purchaser") in the sum of Rs. <<Amount in figures>> (Rupees <<Amount in words>> only) for which payment well and truly to be made to the said Purchaser, the Bank binds itself, its successors and assigns by these presents. Sealed with the Common Seal of the said Bank this <<Date>>

The conditions of this obligation are:

a) If the Bidder having its bid withdrawn during the period of bid validity specified by the Bidder on the Bid Form; or
b) If the Bidder, having been notified of the acceptance of its bid by the Purchaser during the period of validity of bid
    i) Withdraws his participation from the bid during the period of validity of bid document; or
    ii) Fails or refuses to participate in the subsequent Tender process after having been short listed.

We undertake to pay to the Purchaser up to the above amount upon receipt of its first written demand, without the Purchaser having to substantiate its

demand, provided that in its demand the Purchaser will note that the amount claimed by it is due to it owing to the occurrence of one or both of the two conditions, specifying the occurred condition or conditions.

This guarantee will remain in force up to <<insert date>> and including <<extra time over and above mandated in the RFP>> from the last date of submission and any demand in respect thereof should reach the Bank not later than the above date.

NOTHWITHSTANDING ANYTHING CONTAINED HEREIN:

a) Our liability under this Bank Guarantee shall not exceed Rs. <<Amount in figures>> (Rupees <<Amount in words>> only)
b) This Bank Guarantee shall be valid up to <<insert date>>)
c) It is condition of our liability for payment of the guaranteed amount or any part thereof arising under this Bank Guarantee that we receive a valid written claim or demand for payment under this Bank Guarantee on or before <<insert date>>) failing which our liability under the guarantee will automatically cease.

Yours faithfully,

(Authorized Signatory)
Name, Designation & Contact No. Seal

11.1.7 FORM PQ-7: Format for Non-blacklisting Declaration

(To be submitted on the Letterhead of Bidder)

To
The General Manager (Admin),
Odisha Computer Application Centre,
N-1/7-D, Acharya Vihar P.O. RRL, Bhubaneswar - 751013.

Sub: Non-Blacklisting declaration in connection

Madam/Sir,

This is to notify you that our <<Name of the Firm/Company/Organization>> is not declared ineligibility for corrupt and fraudulent practices or blacklisted/debarred by Central Government or any State Government organization / department / PSU in India at the time of submission of the bid.

Yours faithfully,

(Authorized Signatory)
Name, Designation & Contact No.
Seal

11.2 Technical Bid Formats

11.2.1 FORM TECH-1: Description of Proposed Solution

[along with Technology, Scalability, Completeness, Simplicity and Interoperability]

Bidder has to provide details of the entire solution proposed, along with its key differentiators, covering all requirements as listed out in Section 10 of this RFP.

Bidder has to specifically include (but not limited to) diagram and detailed description of the following:

a) Functional Architecture
b) Technical Architecture
c) Network Architecture
d) Deployment Architecture
e) Security Architecture
f) Compliance of specifications for hardware and system software

Bidder must cover all aspects of the solution while showcasing its scalability, completeness, simplicity and interoperability.

11.2.2 FORM TECH-2: Description of Proposed Approach & Methodology

Bidder is free to propose any type of approach for development and implementation

11.2.3 FORM TECH-3: Detailed Work Plan

[with Activities, Duration, Sequencing, Interrelations, Milestones and Dependencies]

SL#	**Deliverable/ Activity***	**Months**							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**n**
a)									
b)									
c)									
d)									
e)									
f)									
g)									
h)									
i)									
j)									
k)									
l)									
m)									
n)									
o)									
p)									
q)									
r)									
s)									

11.2.4 FORM TECH-4: Team Composition, Assignment and Experts' Inputs

No.	Name	Expert's Input (in person/month) per Deliverable (listed in TECH-3)								Total Time-input (in Months)		
		Position	Home/ Field	D-1	D-2	D-3	D-4	.....	D-n	Home	Field	Total
KEY EXPERTS												
K-1			*Home*									
			Field									
K-2			*Home*									
			Field									
K-3			*Home*									
			Field									
K-4			*Home*									
			Field									
K-5			*Home*									
			Field									
NON-KEY EXPERTS (Bidder to determine no. of Non-Key Experts as per its proposed work plan)												
N-1			*Home*									
			Field									
N-2			*Home*									
			Field									
N-n			*Home*									
			Field									
Sub-Total												

11.3 Financial Bid

11.3.1 FORM FIN-1: Financial Bid Covering Letter

(To be submitted on the Letterhead of Bidder)

To

The General Manager (Admin),
Odisha Computer Application Centre,
N-1/7-D, Acharya Vihar P.O. RRL, Bhubaneswar - 751013.

Sub: Selection of Technology Partner For AI-Driven Document Verification Engine under Social Protection Delivery Platform

Madam/Sir,

I /We, the undersigned, offer to provide the service for development and implementation of AI-Driven Document Verification Engine under Social Protection Delivery Platform and our Pre-Qualification, Technical and Financial Proposals. Our attached Financial Proposal is for the sum of <<Amount in words and figures>> exclusive of all applicable taxes and duties.

a) BID PRICE

We declare that our Bid Price is for the entire scope of the work as specified in the in this RFP . These prices are indicated in the Financial Bid as part of this RFP response. In case there is substantial difference between the component wise price approved by OCAC and the price quoted by the bidder, OCAC will have the rights to ask the bidder to realign their prices without impacting the total bid price. We hereby agree to submit our offer accordingly.

b) PERFORMANCE BANK GUARANTEE

We hereby declare that in case the contract is awarded to us, we shall submit the Performance Bank Guarantee as specified in this RFP document.

We understand you are not bound to accept any Proposal you receive. We hereby declare that our Proposal is made in good faith, without collusion or fraud and the information contained in the proposal is true and correct to the best of our knowledge and belief.

We understand that our proposal is binding on us and that you are not bound to accept any proposal you receive.

Yours faithfully,

(Authorized Signatory) Name,
Designation & Contact No. and Seal

11.3.2 FORM FIN-2: Financial Quote

Sl No	Item/Description	Unit	Qty	Unit Cost (Excluding GST)	Total Cost (Excluding GST)
1. Software Development and implementation					
1.1	Development and implementation of AI-driven document verification Application (including development of common APIs as per clause no. 10.9.1)	Lumpsum	1		
1.2	Operation and Management of the entire system for a period of 3 years from the date of Go-live	Year	3		
1.3	Cyber Security Audit of the project by CERT-IN empaneled agency/auditor	Times	4		
1.4	SSL Certificate	Year	3		
1.5	One Support Resource Deployment	Month	36		
1.6	Any third party application/software license/subscription (bidder to define)				
1.7	Cost Discovery item Integration with 3rd party applications in special case (as per clause no. 10.9.2)	Nos.	10		
1.8	Cost Discovery item Machine Learning (AI model training and fine-tuning) for each new document type beyond	Type	5		

	those explicitly specified in this RFP, such as Aadhaar, PAN, etc.				
1.9	Cost Discovery item Software Enhancement Service (Chage Request). Bidder to quote man-month rate for software developer for 20 man month	Man-month	20		
2. AI Infra					
2.1.	Edge AI Runtime Environment	Nos	3		
2.2	Core Hosting Node	Nos	3		
2.3	Setup and Operationalization Costs	Lumpsum			
2.4	Any Other (bidder to specify)		1		
Grand Total Excluding GST					
Total Cost excluding GST in words					

Yours faithfully,

(Authorized Signatory)

Name, Designation & Contact No. Seal